

S3

Benvingut al núvol. En aquest Sprint, faràs el salt de treballar en local (MySQL) a treballar amb **Google BigQuery**, un Data Warehouse serverless dissenyat per a l'anàlisi de dades a gran escala.

En l'entorn empresarial actual, l'analista no acostuma a crear bases de dades des de zero, sinó que ha de connectar-se a repositoris existents (Data Lakes), comprendre els costos del núvol i preparar les dades brutes per a l'anàlisi (ELT).

Lliurament

Revisa el material a entregar en cada exercici.

- **Codi i Infraestructura (.sql)** Els arxius .sql amb tots els scripts generats.
- **Evidència d'Execució i Rendiment (PDF)** Un document PDF únic que contingui les captures de pantalla de la Consola de BigQuery per a cada exercici.
 - **Requisit Crític:** Per validar que la teva optimització funciona, la captura ha de mostrar clarament:
 - El **codi SQL** a l'editor.
 - El missatge de validació (check verd ) o la pestanya Job Information on es vegin els Bytes processats (Bytes processed).

Important

Recorda revisar les [millors pràctiques](#) abans de fer el lliurament.

- **Control de Costos:** A BigQuery, cada consulta costa diners. Abans d'executar res, fixa't sempre en el missatge del validador (Dry Run) a dalt a la dreta.
- **Ubicació de les Dades:** Assegura't que el teu Dataset estigui a la regió EU (Multiple regions) o les connexions fallaran.

Limitacions del Sandbox

La zona de proves de BigQuery (Sandbox) està subjecta als següents límits:

- S'apliquen tots els límits i quotes habituals de BigQuery.
- Tens 10 GB d'emmagatzematge actiu i 1 TB de dades de consulta processades cada mes sense cap cost.
-  **Caducitat:** Totes les taules, vistes i particions caduquen de forma automàtica després de **60 dies**.
- No admet funcions avançades com: Transmissió de dades (Streaming), DML (UPDATE / DELETE massius) o el Servei de transferència de dades.

Nivell 1: Entorn i Ingesta Híbrida (Code-First)

Escenari:

Acabeu d'arribar a l'empresa. L'equip d'Enginyeria de Dades ha deixat els fitxers històrics al Data Lake corporatiu.

La teva primera missió és dissenyar i crear l'entorn de treball. El CTO exigeix professionalitat: no abocarem dades sense ordre, sinó que implementarem una Arquitectura Medallion.

Exercici 1: Arquitectura de Dades (Lògica vs. Física)

Abans de tocar la consola, has d'entendre la diferència entre allò que dissenyem i allò que construïm:

- **Arquitectura Lògica (Capes de Dades):** És el disseny conceptual. Organitza les dades segons el seu nivell de qualitat (Bronze, Silver, Gold).
- **Implementació Física (Datasets):** És el recurs real a Google Cloud. A BigQuery, utilitzarem Datasets (contenidors) per materialitzar físicament aquestes capes lògiques.



Tasques:

Implementareu físicament l'arquitectura lògica creant tres Datasets dins d'un mateix projecte per centralitzar la facturació i els permisos.



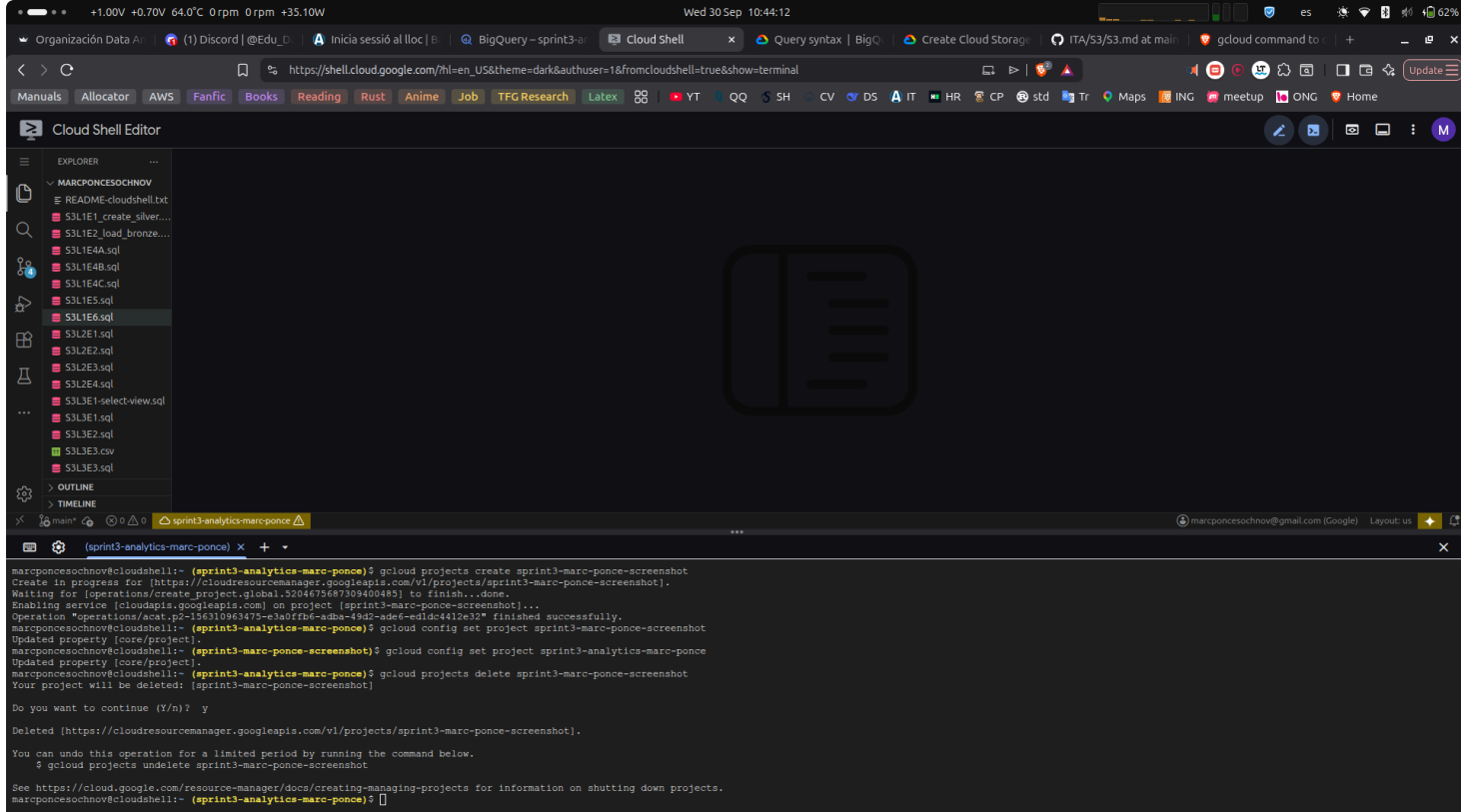
Crea un **Nou Projecte** a Google Cloud anomenat:

```
sprint3-analytics-[el-teu-nom] .
```

Ho pots crear per la UI, però si ho fas per Consola (gcloud CLI) té un plus.

```
gcloud projects create sprint3-analytics-marc-ponce  
gcloud config set project sprint3-analytics-marc-ponce
```

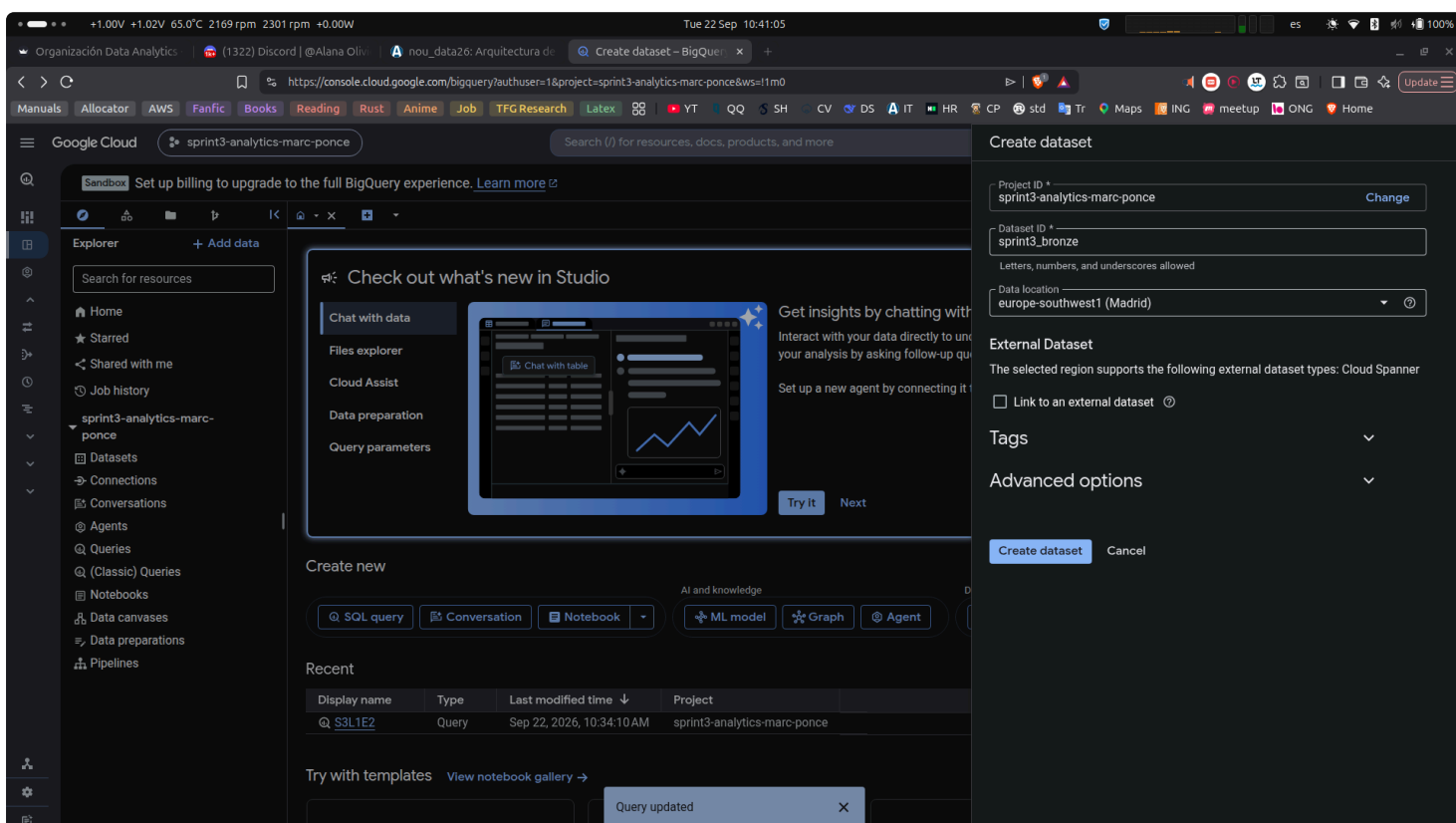
» I forgot to screenshot the execution of the command. So I created and deleted a project for the sake of having a screenshot:



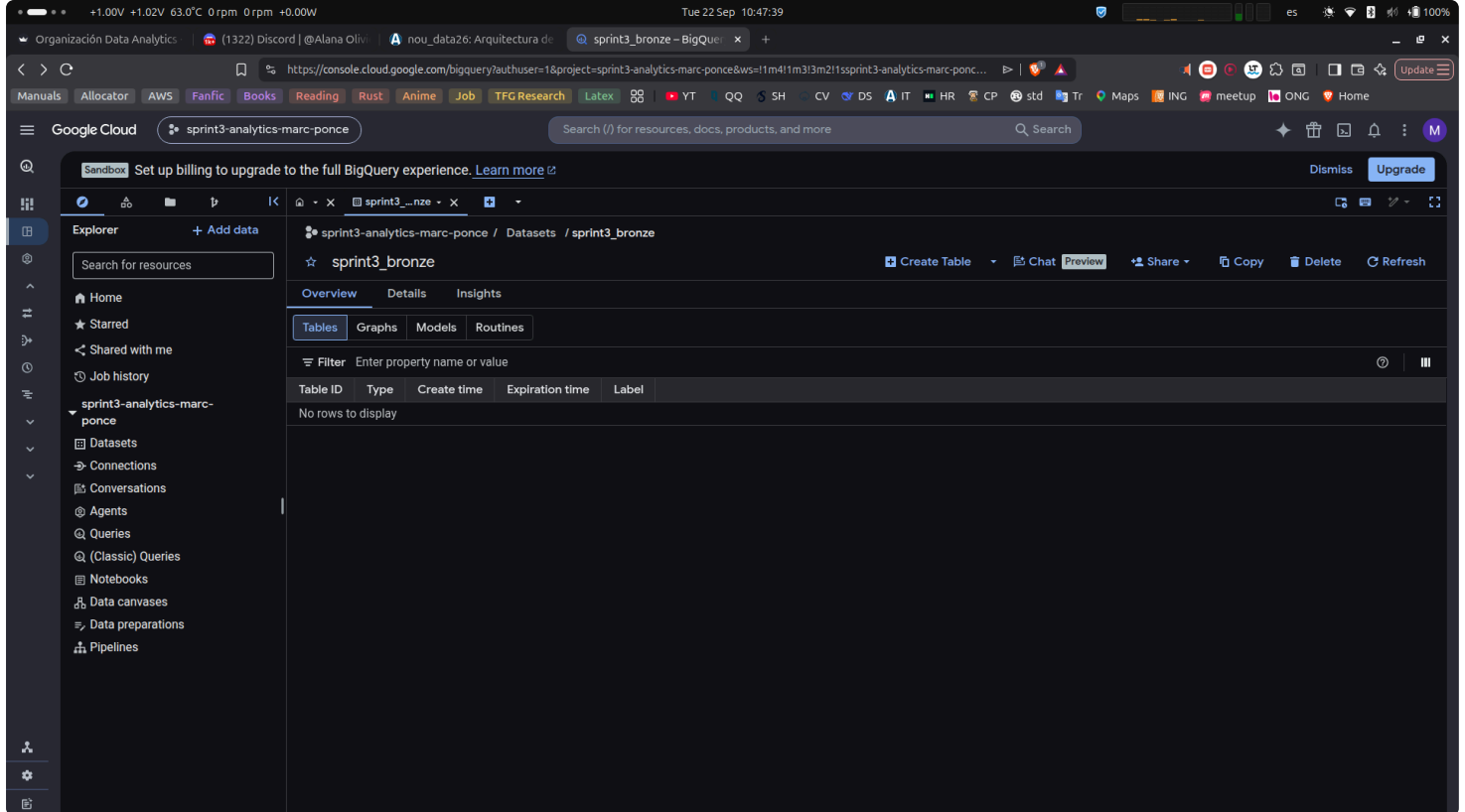
Per demostrar el teu domini total de la plataforma, crearàs cadascun dels datasets utilitzant un mètode diferent:

- **Dataset Físic** `sprint3_bronze` → Capa Lògica Bronze (Dades Crues)
 - Funció: Dades crues tal com arriben de la font.
 - Mètode: Utilitza la **Interfície Gràfica (UI)** de BigQuery (fent clics).

» Click the three points icon in **Datasets** on the left panel, then **Create dataset**. Input the name of the **SCHEMA**, and its location to an EU server: `europa-southwest1` (Madrid).



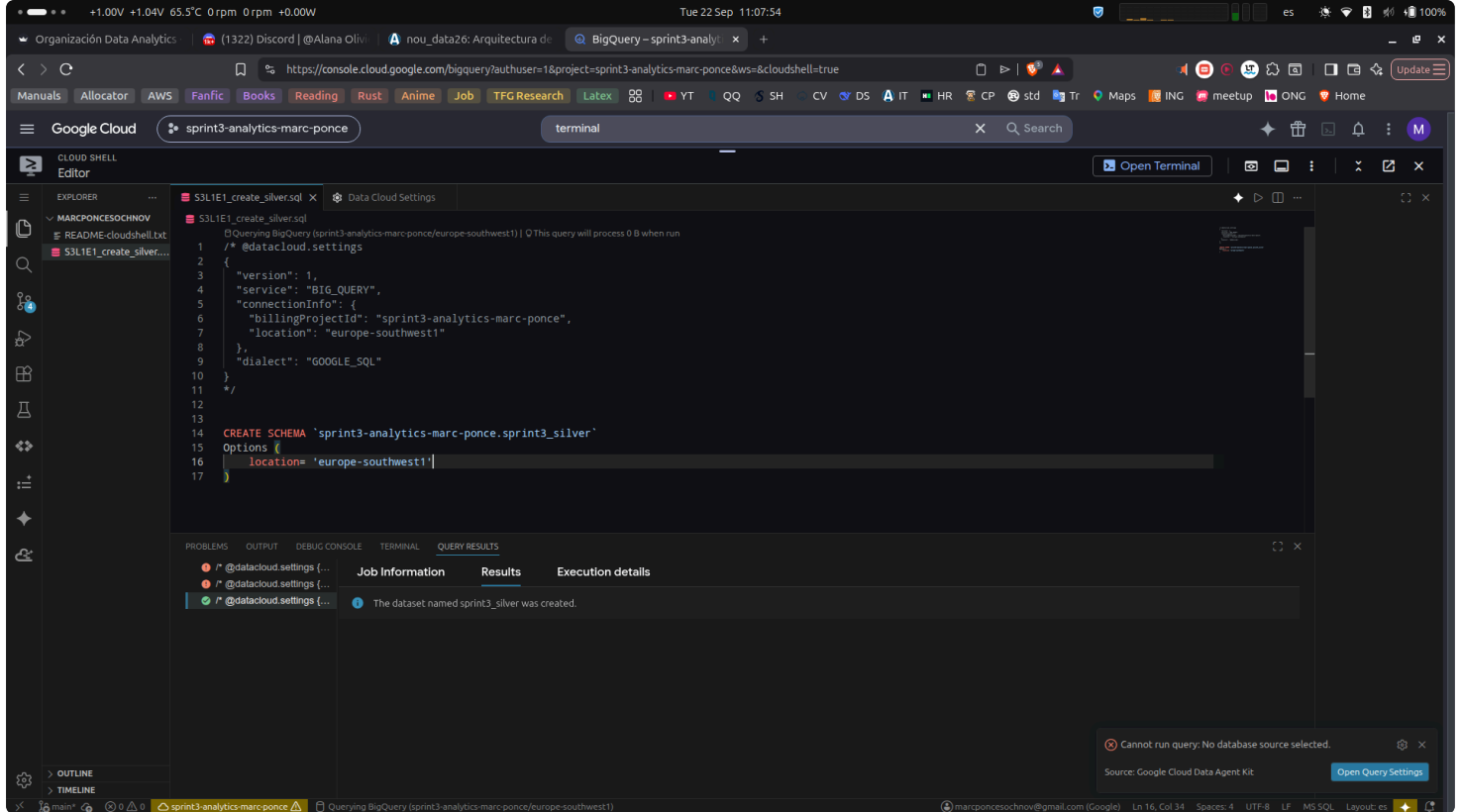
» The **SCHEMA** was created successfully:



- **Dataset Físic** sprint3_silver → Capa Lògica Silver (Dades Netes)
 - Funció: Dades netes, tipades i de-duplicades.
 - Mètode: Utilitza codi SQL (CREATE SCHEMA).

```
/* @datacloud.settings
{
  "version": 1,
  "service": "BIG_QUERY",
  "connectionInfo": {
    "billingProjectId": "sprint3-analytics-marc-ponce",
    "location": "europe-southwest1"
  },
  "dialect": "GOOGLE_SQL"
}
*/

CREATE SCHEMA `sprint3-analytics-marc-ponce.sprint3_silver`
Options (
  location= 'europe-southwest1'
)
```

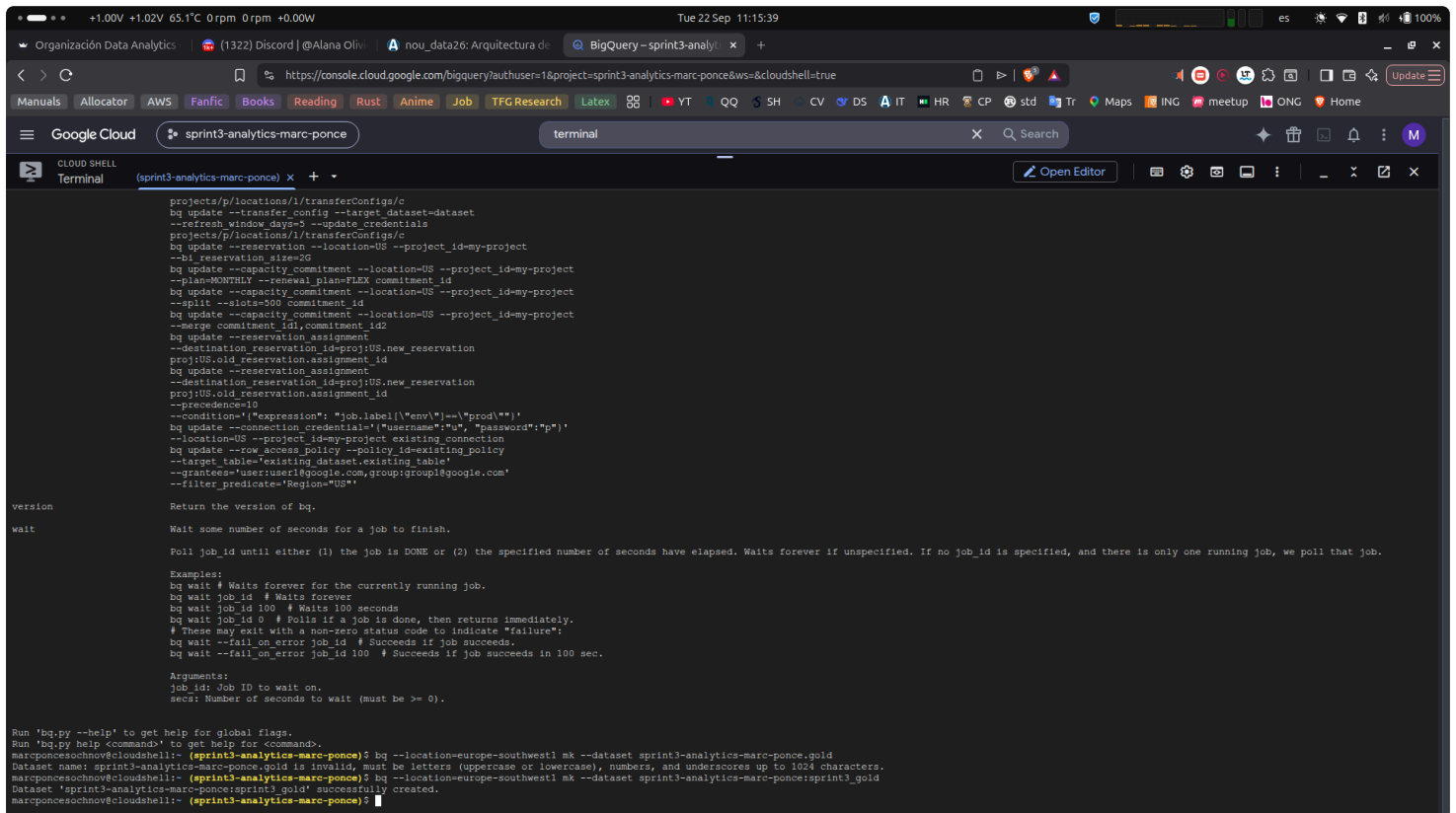


- **Dataset Físic** sprint3_gold → Capa Lògica Gold (Dades de Negoci)
 - Funció: Dades agregades llestes per a informes i panells de control (dashboards).
 - Mètode: Utilitza Cloud Shell (Línia d'ordres bq).

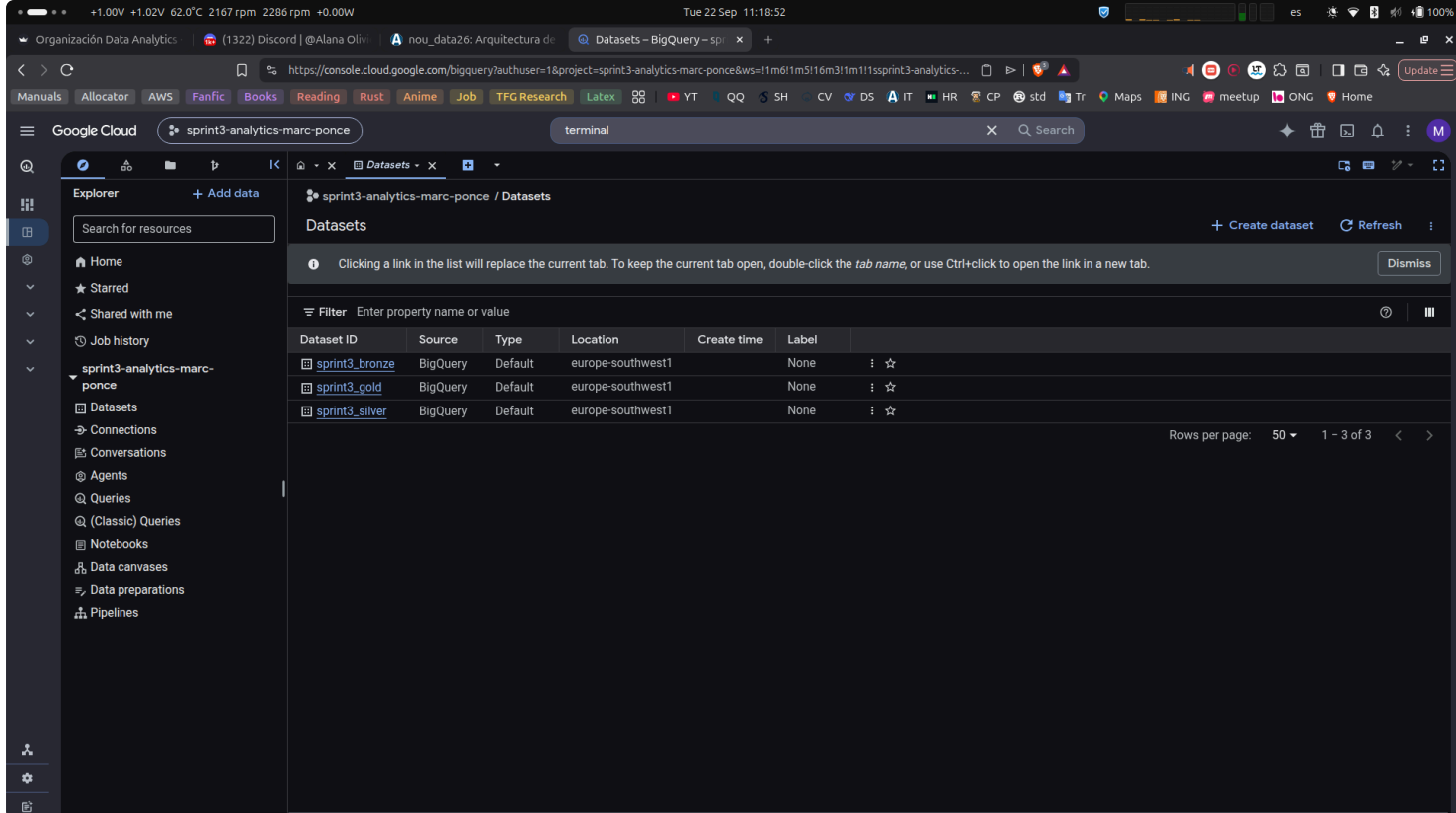
```

bq --location=europe-southwest1 \
mk --dataset sprint3-analytics-marc-ponce:sprint3_gold

```



» In this last screenshot you can confirm that the datasets have been created successfully:



Tingueu en compte la ubicació de les dades originals per decidir la ubicació física dels tres Datasets.

 **Nota:** Encara que siguin capes lògiques diferents, físicament han de viure a la mateixa regió per poder "parlar" entre elles sense costos de transferència.

Exercici 2: Ingesta en Capa Bronze (Connexió DDL)

Un analista professional automatitza les tasques. Ara que tens la teva arquitectura a punt, poblarem la primera capa (**Bronze**) sense moure les dades originals, creant **Taules Externes** que apuntin al Data Lake.

Tasca:

Escriu i executa les sentències `CREATE EXTERNAL TABLE` per connectar els següents fitxers al `dataset sprint3_bronze`. Para molta atenció a les "Notes Tècniques", ja que no tots els arxius tenen el mateix format.

Taula	Domini	Ruta al Núvol (URI)	Notes Tècniques
transactions_raw	ERP	transactions.csv	Delimitador: ;
companies_raw	ERP	companies.csv	Capçalera: Ignora la 1a fila. Defineix l'esquema manualment perquè tot és text.
american_users_raw	CRM	american_users.csv	Estàndard.
european_users_raw	CRM	european_users.csv	Estàndard.
credit_cards_raw	CRM	credit_cards.csv	Estàndard.

Entrega:

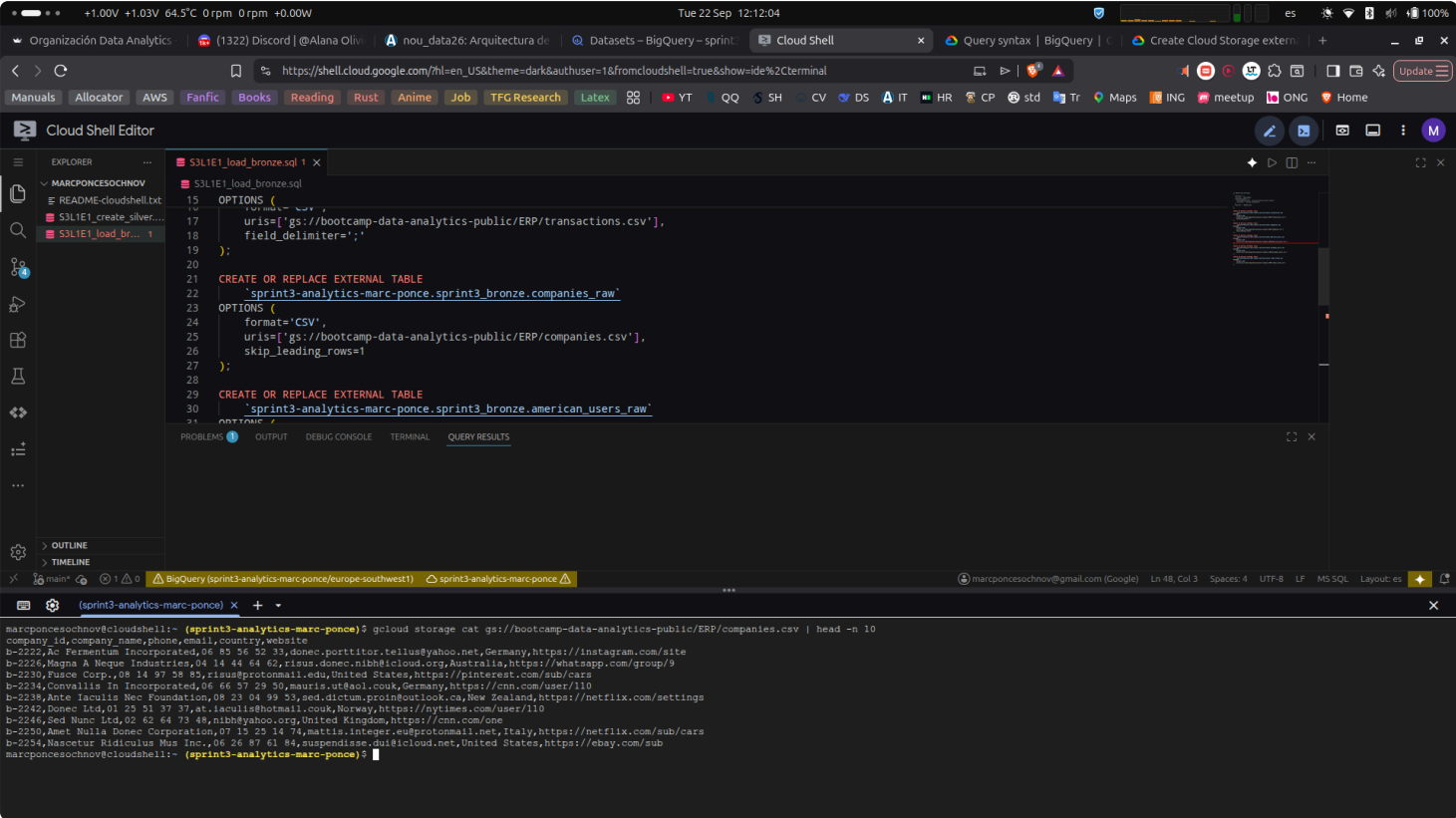
El codi SQL utilitzat per a `transactions_raw` i una captura de pantalla amb les 5 taules creades.

» To understand the structure of `companies.csv`, we `cat | head` the file in the terminal:

```
gcloud storage cat gs://bootcamp-data-analytics-public/ERP/companies.csv | head -n 10
```

» Here is the resulting table:

company_id	company_name	phone	email	country	website
b-2222	Ac Fermentum Incorporated	06 85 56 52 33	donec.porttitor.tellus@yahoo.net	Germany	https://insta
b-2226	Magna A Neque Industries	04 14 44 64 62	risus.donec.nibh@icloud.org	Australia	https://wha



» Next I will provide the code, and a screenshot of its successful execution:

```
CREATE OR REPLACE EXTERNAL TABLE
`sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw`
OPTIONS (
  format='CSV',
  uris=['gs://bootcamp-data-analytics-public/ERP/transactions.csv'],
  field_delimiter=';';
);

CREATE OR REPLACE EXTERNAL TABLE
`sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw`
(
  company_id STRING,
  company_name STRING,
```

```

    phone STRING,
    email STRING,
    country STRING,
    website STRING
)
OPTIONS (
    format='CSV',
    uris=['gs://bootcamp-data-analytics-public/ERP/companies.csv'],
    skip_leading_rows=1
);

```

CREATE OR REPLACE EXTERNAL TABLE

```

`sprint3-analytics-marc-ponce.sprint3_bronze.american_users_raw`
OPTIONS (
    format='CSV',
    uris=['gs://bootcamp-data-analytics-public/CRM/american_users.csv']
);

```

CREATE OR REPLACE EXTERNAL TABLE

```

`sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw`
OPTIONS (
    format='CSV',
    uris=['gs://bootcamp-data-analytics-public/CRM/european_users.csv']
);

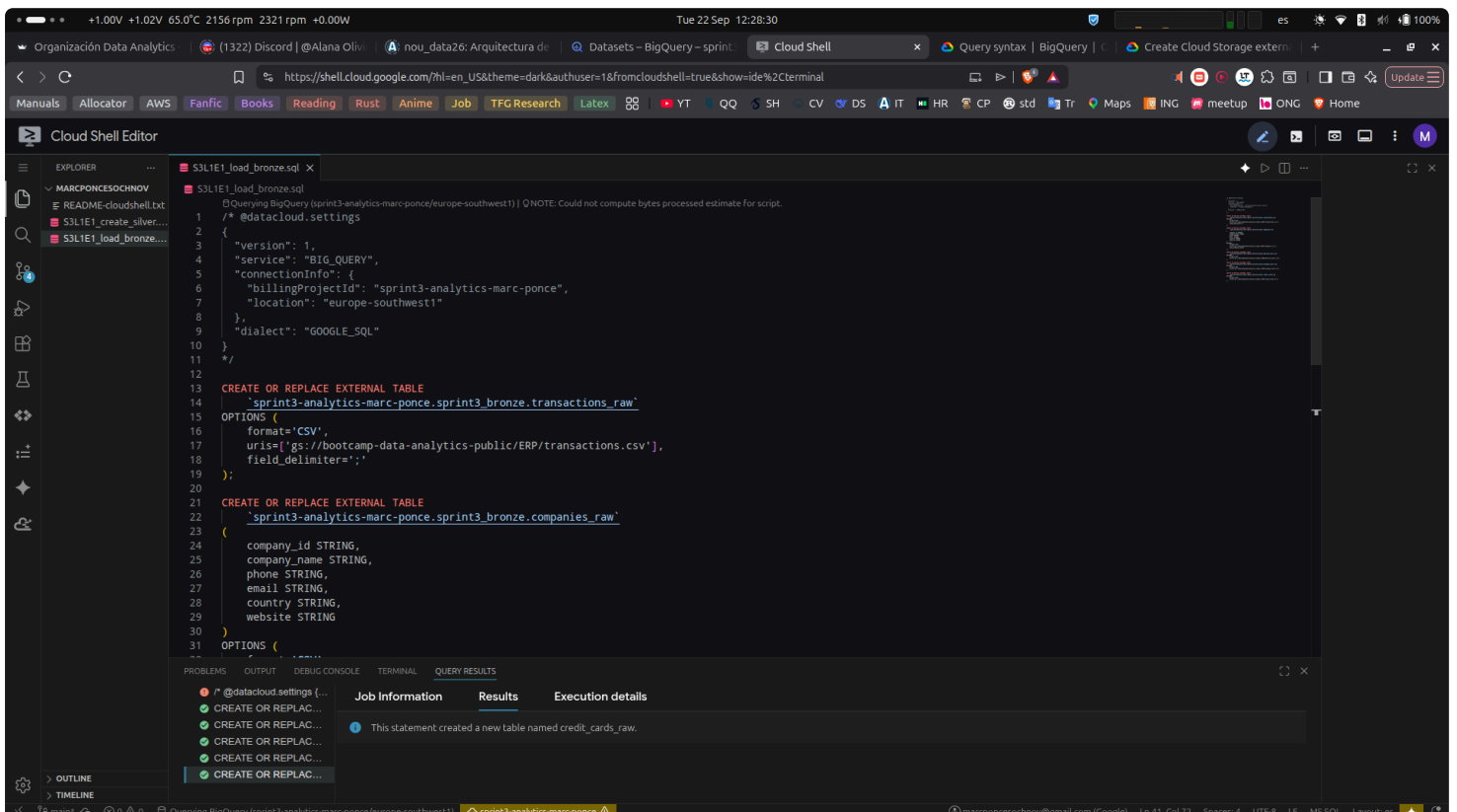
```

CREATE OR REPLACE EXTERNAL TABLE

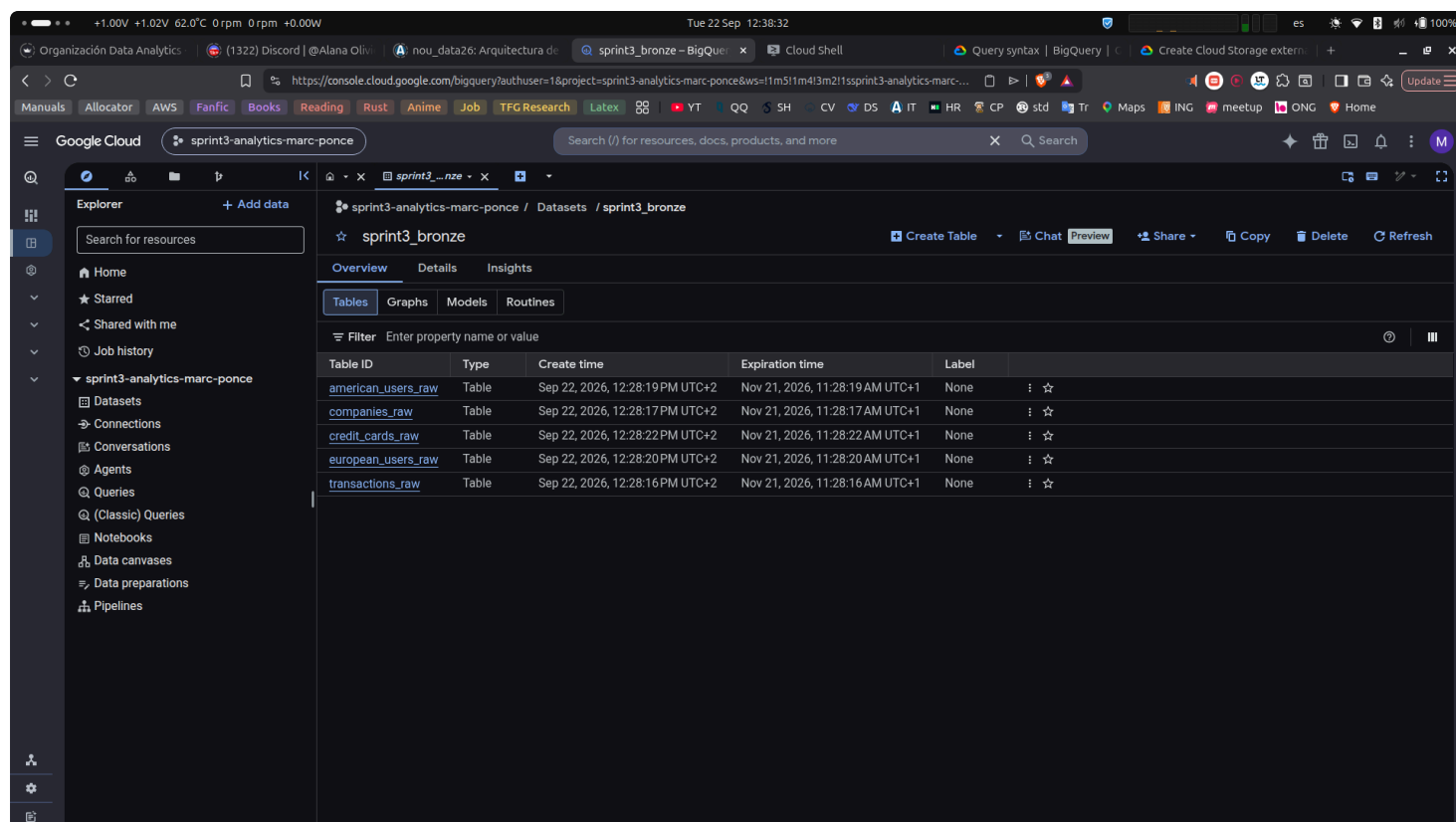
```

`sprint3-analytics-marc-ponce.sprint3_bronze.credit_cards_raw`
OPTIONS (
    format='CSV',
    uris=['gs://bootcamp-data-analytics-public/CRM/credit_cards.csv']
);

```



» Lastly we check that the tables are in place:



The screenshot shows the Google Cloud BigQuery console interface. The left sidebar displays the 'Explorer' panel with the 'sprint3-analytics-marc-ponce' project selected. The main panel shows the 'sprint3_bronze' dataset with a list of tables. The tables listed are:

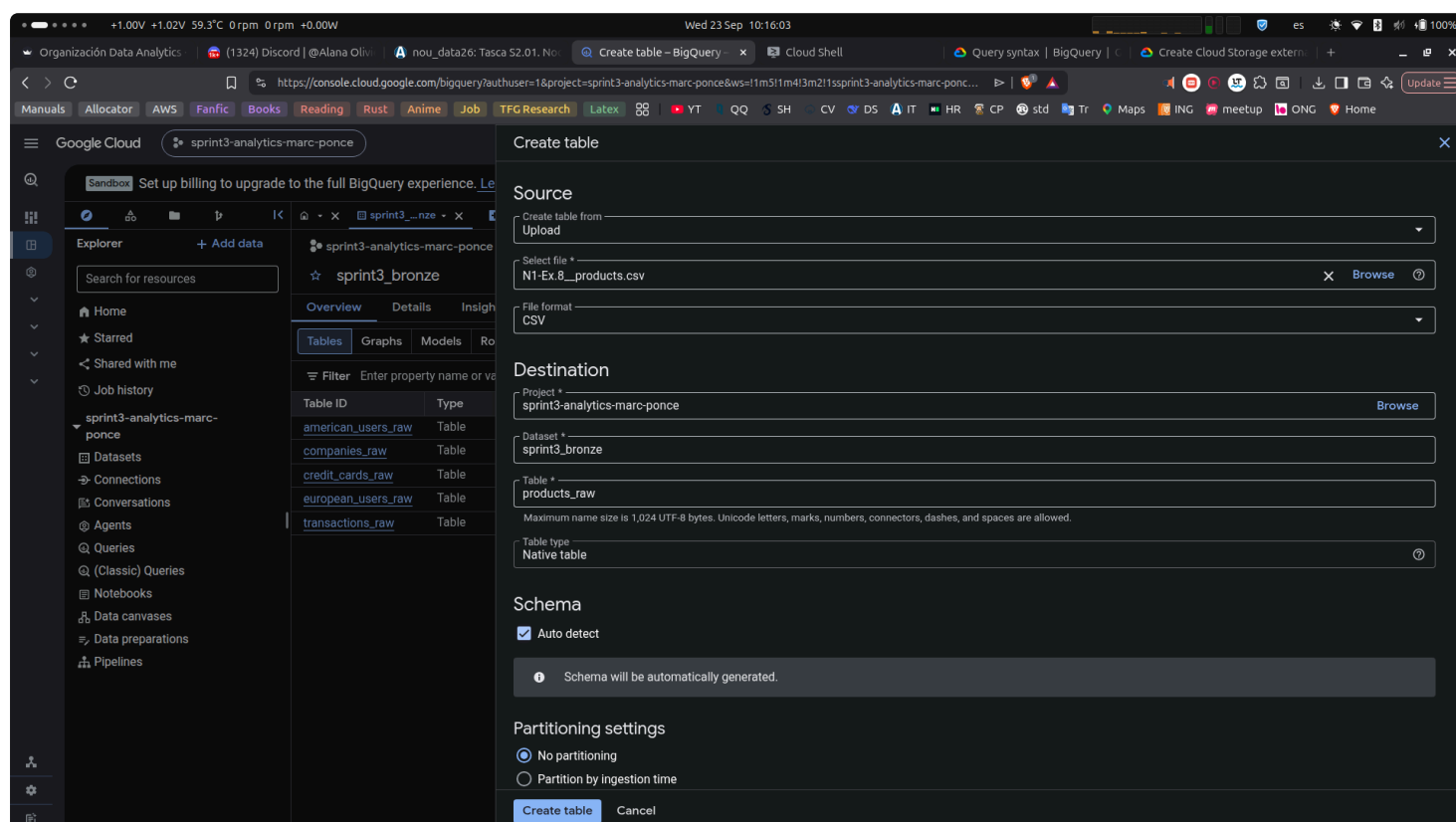
Table ID	Type	Create time	Expiration time	Label
american_users_raw	Table	Sep 22, 2026, 12:28:19 PM UTC+2	Nov 21, 2026, 11:28:19 AM UTC+1	None
companies_raw	Table	Sep 22, 2026, 12:28:17 PM UTC+2	Nov 21, 2026, 11:28:17 AM UTC+1	None
credit_cards_raw	Table	Sep 22, 2026, 12:28:22 PM UTC+2	Nov 21, 2026, 11:28:22 AM UTC+1	None
euopean_users_raw	Table	Sep 22, 2026, 12:28:20 PM UTC+2	Nov 21, 2026, 11:28:20 AM UTC+1	None
transactions_raw	Table	Sep 22, 2026, 12:28:16 PM UTC+2	Nov 21, 2026, 11:28:16 AM UTC+1	None

Exercici 3: Càrrega de Dades Locals (Upload)

Falta el catàleg de **Productes**. Aquesta informació no és al Data Lake, utilitza el fitxer *products.csv* de l'Sprint 2.

- Carrega aquest fitxer manualment ("Upload") al dataset `sprint3_bronze` i crea la taula `products_raw`.
- Observa que aquesta serà l'única **Taula Nativa** de moment.

» Upload the `.csv` to `sprint3_bronze.products_raw`

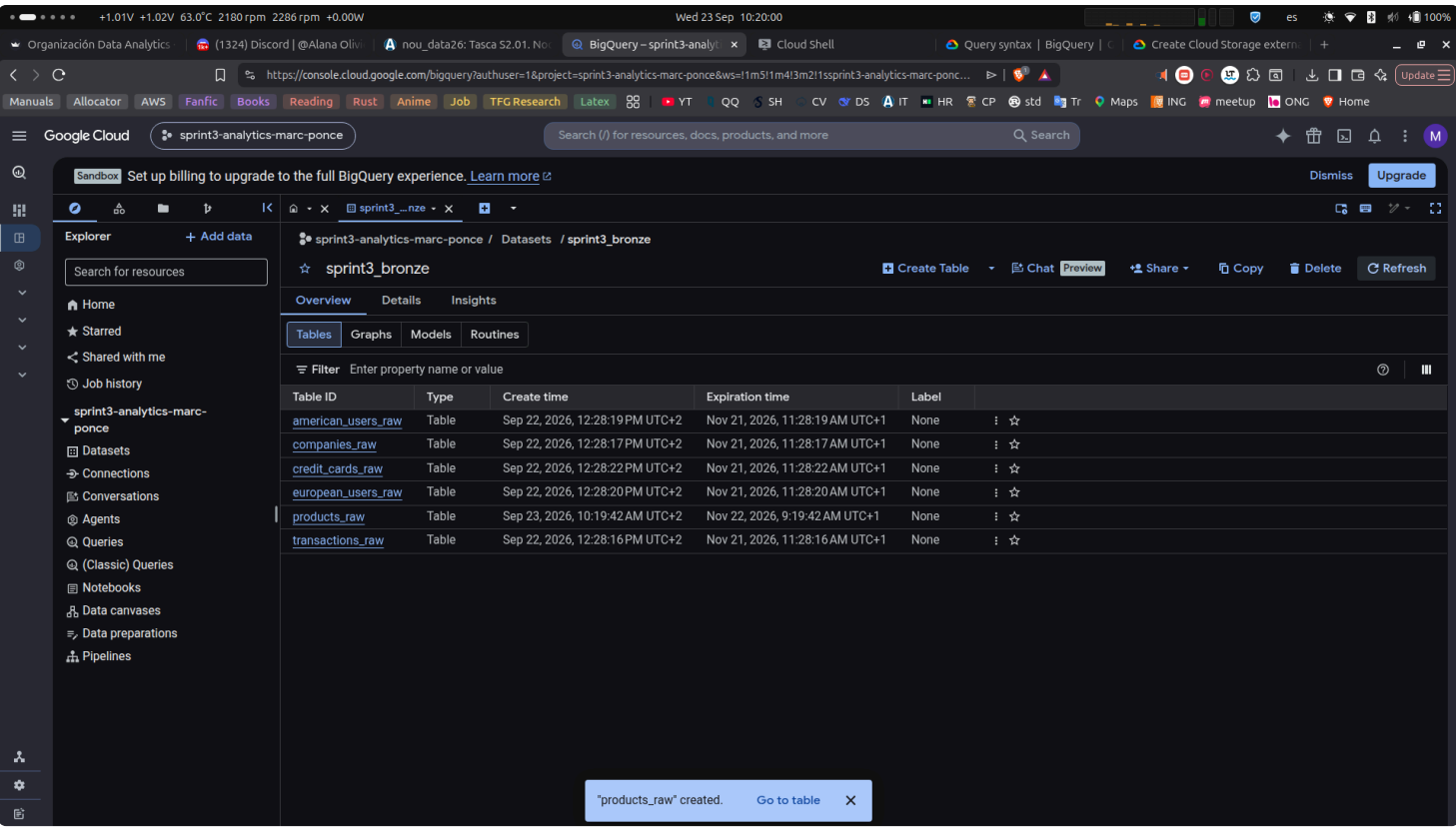


The screenshot shows the Google Cloud BigQuery console interface with the 'Create table' dialog box open. The dialog box has the following fields and options:

- Source:** 'Create table from' is set to 'Upload'. 'Select file' is 'N1-Ex.8__products.csv'.
- File format:** 'CSV'.
- Destination:** 'Project' is 'sprint3-analytics-marc-ponce'. 'Dataset' is 'sprint3_bronze'. 'Table' is 'products_raw'.
- Table type:** 'Native table'.
- Schema:** 'Auto detect' is checked. A message states: 'Schema will be automatically generated.'
- Partitioning settings:** 'No partitioning' is selected.

The 'Create table' button is highlighted in blue.

» Confirm its presence in `sprint3_bronze`.



Exercici 4: Arquitectura i Rendiment. Materialització de Dades (Assistit per IA)

Escenari:

Les consultes sobre el Data Lake van lentes. Has de demostrar al teu mànager la diferència de rendiment entre treballar amb fitxers externs i treballar amb taules natives de BigQuery.

a) Materialització de Dades (Assistit per IA)

Tasca:

Crea una nova taula anomenada `sprint3_bronze.transactions_raw_native` que sigui una còpia exacta de la teva taula externa `transactions_raw`.

Requisit de "AI-Pair Programming":

No escriguis el codi SQL manualment. Utilitza l'Assistent d'IA (Gemini) integrat a BigQuery (o el xat lateral) per generar la sentència.

👉 **El Teu Rol:** Ets el responsable de validar que el codi que et dona la IA és correcte abans d'executar-lo.

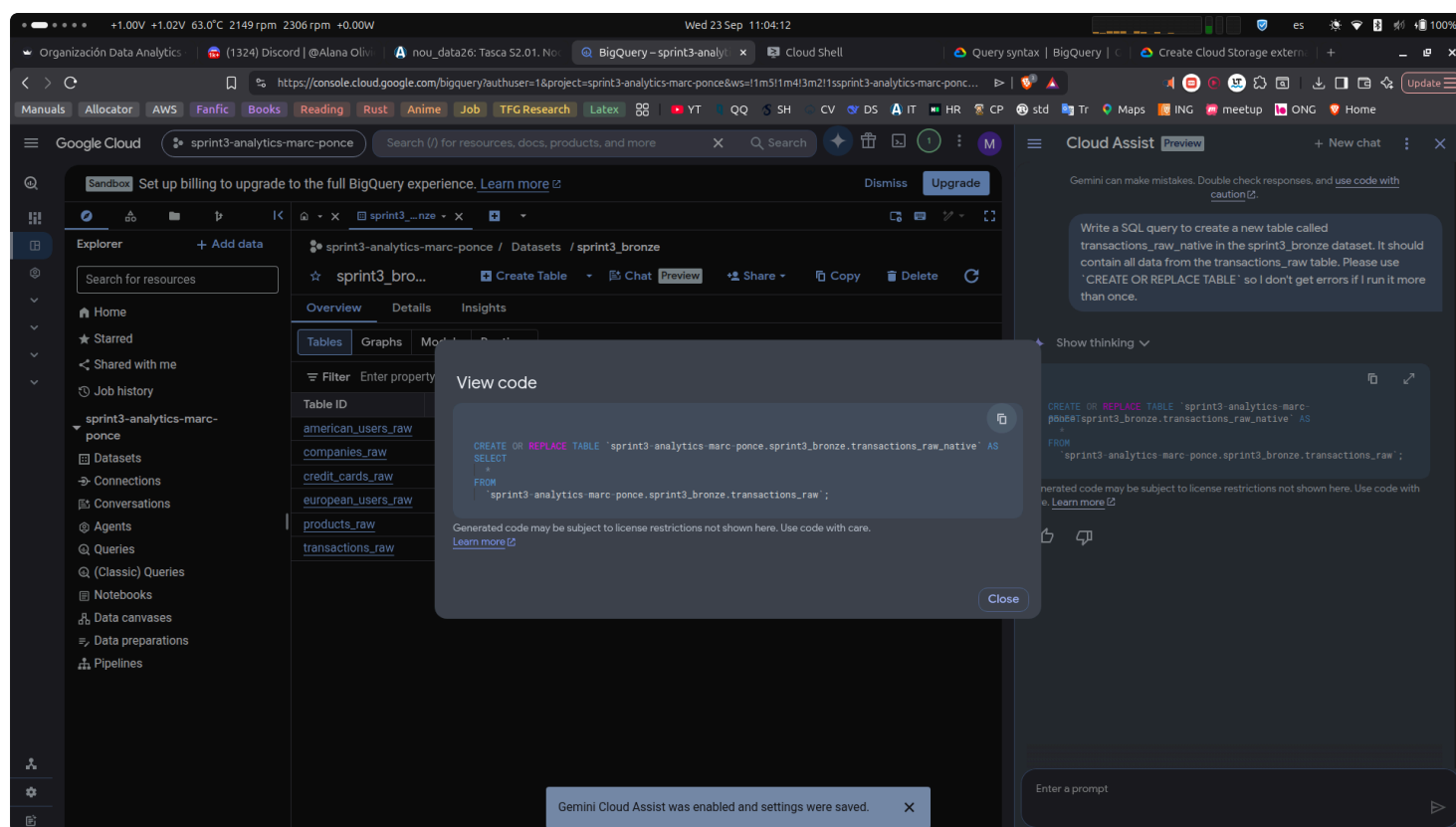
Prompt Suggestit:

"Write a SQL query to create a new table called `transactions_raw_native` in the `sprint3_bronze` dataset. It should contain all data from the `transactions_raw` table. Please use `CREATE OR REPLACE TABLE` so I don't get errors if I run it more than once."

👉 Entrega:

Una captura de pantalla on es veigi:

- El prompt que vas escriure a la IA.

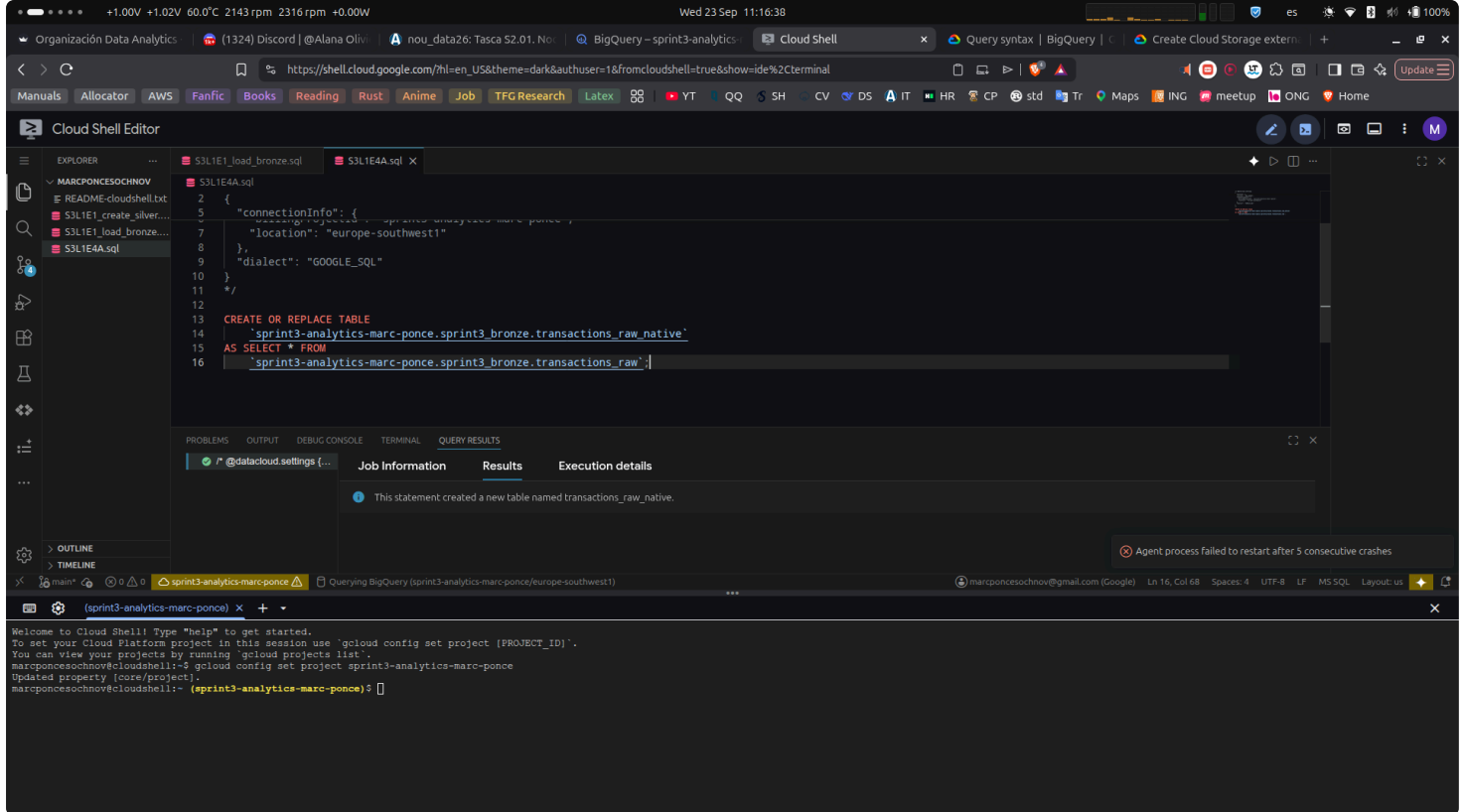


- El codi SQL que la IA va generar.

```
CREATE OR REPLACE TABLE
`sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native`
AS SELECT * FROM
`sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw`;
```

» So far I can see the code makes sense: we are just creating a table that is not `EXTERNAL`, which I am quite sure means it's `native`. Created from a standard `SELECT *`, the names are not mangled, so I deem this code correct.

- El resultat de l'execució ("This statement created a new table...").



» "This statement created a new table named `transactions_raw_native`."

b) Auditoria de Costos

Tasca:

Dissenyar una prova per comparar el cost de llegir una sola columna (ex: `id`) a la taula externa vs. la taula nativa. Utilitza els Bytes processats i els Bytes facturats per fer la comparació.

```
SELECT t.id
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw` AS t;

SELECT t.id
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t;
```

- Quan "pesa" la consulta a `transactions_raw` ?

Cloud Shell Editor

```
2 {
3   "dialect": "GOOGLE_SQL"
4 }
5 */
6
7 SELECT t.id
8 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw` AS t;
9
10 SELECT t.id
11 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t;
```

Job Information

Job ID	sprint3-analytics-marc-ponce:europa-southwest1.script_job_77eb85f86e4a574b55d2f41c69a23475_0
User	MarcPonceSochnov@gmail.com
Location	europa-southwest1
Creation time	Sep 23, 2026, 11:31:16AM
Start time	Sep 23, 2026, 11:31:16AM
End time	Sep 23, 2026, 11:31:17AM
Duration	1.194 sec
Bytes processed	13220706
Bytes billed	13631488
Slot milliseconds	1398
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce_5344bfc8b113c946a01148977895b2373c7bdc09.anon0c973f1bd4c38755f688fd574d8d4d6871e246eae3af1d643d98bbf7d7cac223
Query	SELECT t.id FROM `sprint3-analytics-marc-ponce.sprint3_`
Labels	

Processed	Billed
13.220 MB	13.631 MB

- Quan "pesa" la mateixa consulta a `transactions_raw_native` ?

Cloud Shell Editor

```
2 {
3   "dialect": "GOOGLE_SQL"
4 }
5 */
6
7 SELECT t.id
8 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw` AS t;
9
10 SELECT t.id
11 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t;
```

Job Information

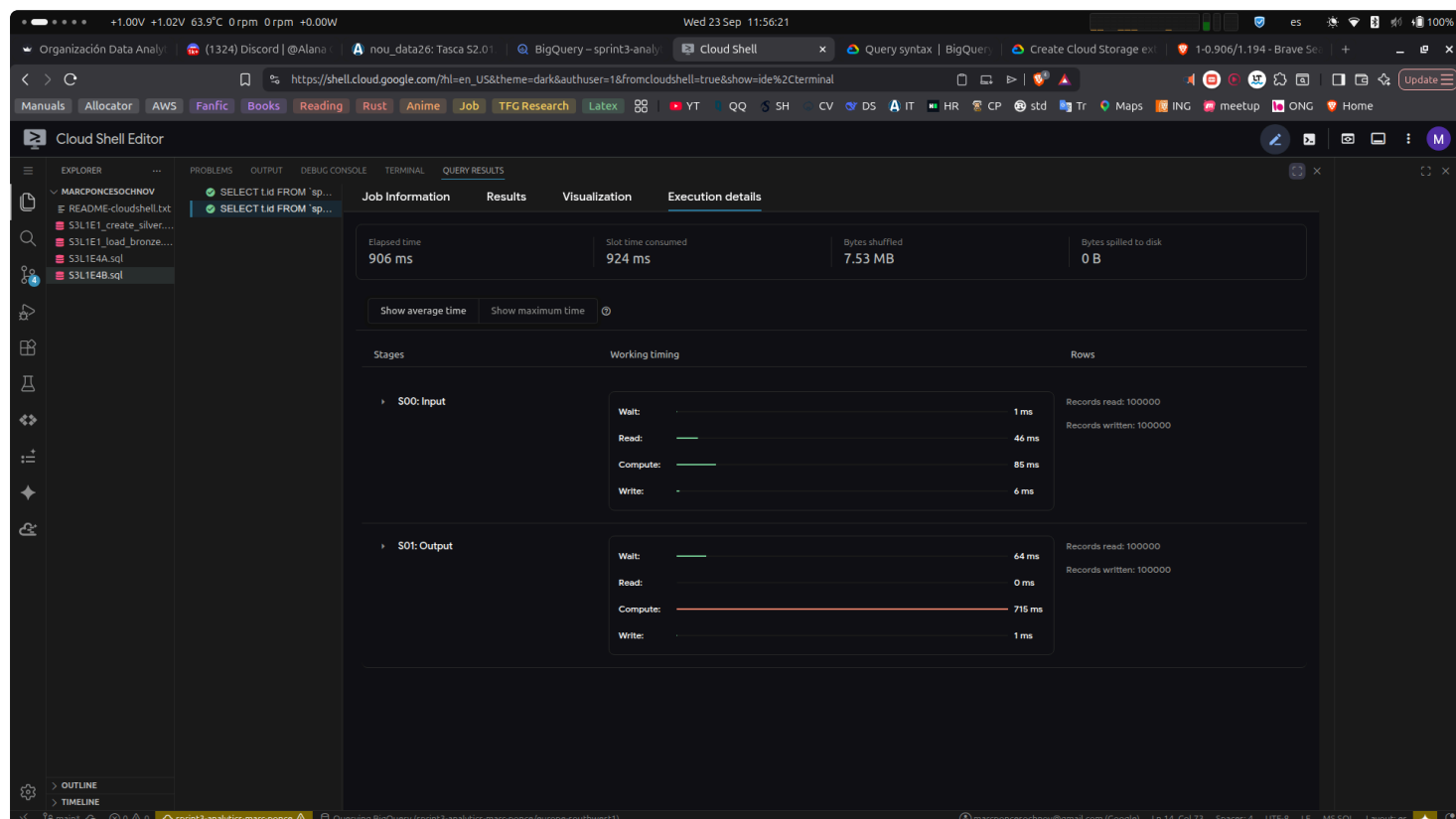
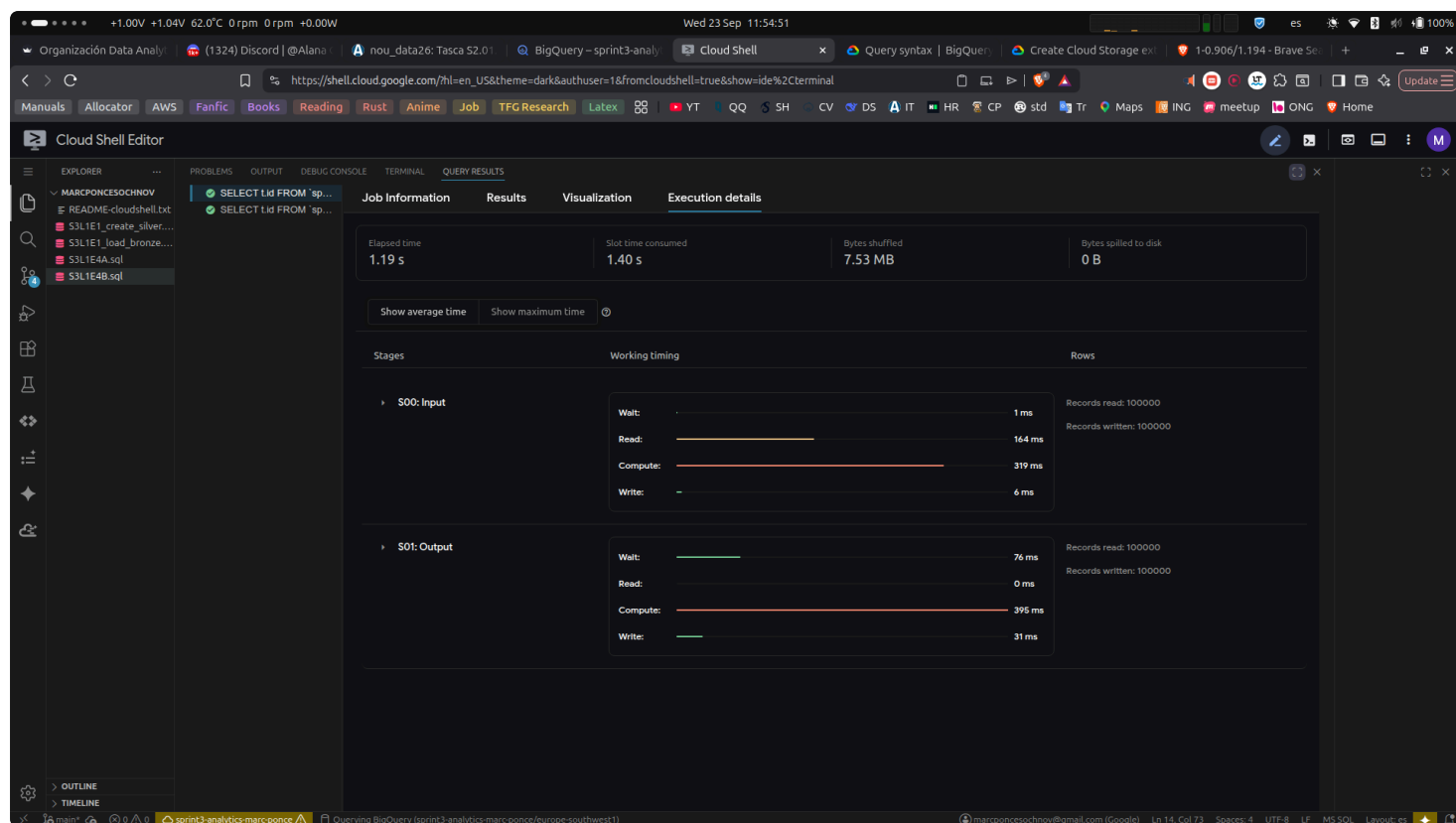
Job ID	sprint3-analytics-marc-ponce:europa-southwest1.script_job_ba4452cee0b4e8244baf1dd1678248c_1
User	MarcPonceSochnov@gmail.com
Location	europa-southwest1
Creation time	Sep 23, 2026, 11:31:18AM
Start time	Sep 23, 2026, 11:31:18AM
End time	Sep 23, 2026, 11:31:19AM
Duration	0.906 sec
Bytes processed	3800000
Bytes billed	10485760
Slot milliseconds	924
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce_5344bfc8b113c946a01148977895b2373c7bdc09.anon8a6b6b1d496440ce2bf74236e5d783108092940d751cc6369134530412b85995
Query	SELECT t.id FROM `sprint3-analytics-marc-ponce.sprint3_`
Labels	

Processed	Billed
3.800 MB	10.485 MB

Entrega:

Captures del validador mostrant la diferència i una conclusió tècnica.

» The difference in billing is of 3MB , which makes the native table 23% cheaper. Comparing the execution times, it is also 24% faster.



» We can also observe that the compute and read in S00 varies much more than some 20-25% , which might indicate a great difference in energy consumption.

c) El perill del LIMIT

El LIMIT estalvia dades processades a BIGQUERY?



Fes un **LIMIT 10** sobre la taula `transactions_raw_native` y la taula `transactions_raw`

```
SELECT t.id
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw` AS t
LIMIT 10;

SELECT t.id
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
LIMIT 10;
```

✅ **Comprovació:** Mira el "Dry Run". Ha baixat el cost respecte a no posar **LIMIT** ? Explica breument per què això és una trampa per a principiants.

The screenshot shows the Google Cloud Shell Editor interface. The left sidebar contains an Explorer view with a file tree for 'MARCPONCESOCHNOV'. The main editor area displays two SQL queries. The first query selects from 'transactions_raw' with a LIMIT of 10. The second query selects from 'transactions_raw_native' with a LIMIT of 10. Below the editor, the 'QUERY RESULTS' tab is active, showing the output of a dry run command: 'bq query --dry_run < S3L1E4C.sql'. The output states: 'Query successfully validated. Assuming the tables are not modified, running this query will process lower bound of 3800000 bytes of data.'

» The dry run yields a lower bound of 3.8MB. Lower bound means the minimum amount of bytes that it expects will be processed. There is no upper bound. I have to mention that it doesn't seem to account for the `EXTERNAL TABLE` query, and the `LIMIT` does not seem to cut off the bytes that it expects to process.

Cloud Shell Editor interface showing a query execution job. The query is `SELECT t.id FROM `sprint3-analytics-marc-ponce.sprint3_...`. The job information is as follows:

Job ID	sprint3-analytics-marc-ponce:europa-southwest1_script_job_01689bb76aa80f727b5a351b2c68655_0
User	MarcPonceSochnov@gmail.com
Location	europa-southwest1
Creation time	Sep 23, 2026, 12:17:51 PM
Start time	Sep 23, 2026, 12:17:52 PM
End time	Sep 23, 2026, 12:17:52 PM
Duration	0.537 sec
Bytes processed	1471
Bytes billed	10485760
Slot milliseconds	204
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce_53a4bfc8b113c946a01148977895b2373c7bdc09.anona4351b0e32edac00875cec689d572c50783294ca98d0872f98025ad7ed34797a
Query	SELECT t.id FROM `sprint3-analytics-marc-ponce.sprint3_...
Labels	

Cloud Shell Editor interface showing the execution details of the query. The execution details are as follows:

Elapsed time	Slot time consumed	Bytes shuffled	Bytes spilled to disk
537 ms	204 ms	410 B	0 B

Stages: S00: Output

Working timing	Rows
Wait: 1 ms	Records read: 11
Read: 119 ms	Records written: 10
Compute: 20 ms	
Write: 3 ms	

» With `transactions_raw` we can observe that the `LIMIT` did cut the amount of bytes processed. In fact, it can be further seen that only 11 records were processed. But we were billed 10MB regardless.

The screenshot shows the Google Cloud Shell Editor interface. On the left is the Explorer pane with a file tree containing 'MARCPONCESOCHNOV', 'README-cloudshell.txt', and several SQL files. The main area displays the 'Job Information' tab for a BigQuery job. The job details are as follows:

Job ID	sprint3-analytics-marc-ponce-europe-southwest1_script_job_7b8a0456f122b4f13ee3b9c703d435c1_1
User	MarcPonceSochnov@gmail.com
Location	europe-southwest1
Creation time	Sep 23, 2026, 12:17:53 PM
Start time	Sep 23, 2026, 12:17:53 PM
End time	Sep 23, 2026, 12:17:53 PM
Duration	0.161 sec
Bytes processed	3800000
Bytes billed	10485760
Slot milliseconds	43
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce_53a4bfc8b113c946a01148977895b2373c7bdc09anon2e96c3bb43c252c0f6e2a22518a73fda17ce4dac4bde398c16790e9a961813a
Query	SELECT t.id FROM `sprint3-analytics-marc-ponce.sprint3_...`
Labels	

A status message at the bottom right indicates: "A query is already running in this editor."

This screenshot shows the 'Execution details' tab for the same BigQuery job. It provides a breakdown of the execution performance:

Metric	Value
Elapsed time	161 ms
Slot time consumed	43 ms
Bytes shuffled	410 B
Bytes spilled to disk	0 B

Below these metrics, there are sections for 'Stages' and 'Working timing'. The 'S00: Output' stage is expanded, showing a bar chart of its internal operations:

Operation	Time
Wait	1 ms
Read	10 ms
Compute	18 ms
Write	3 ms

Additional statistics for the stage are listed on the right: "Records read: 100000" and "Records written: 10". A status message at the bottom right indicates: "A query is already running in this editor."

» With `transactions_raw_native` we are billed the same 10MB . It is faster, and most likely uses less energy. The amount of bytes processed doesn't differ from the `dry-run` lower bound estimation, and it reads and processes all 10000 records.

» `LIMIT` does make the numbers change, but it does not pause the machine after reading the first ten rows. Consulting an AI search engine I have come to the following information:

“Although the `LIMIT` clause limits the rows that a query produces, it doesn't limit the amount of data processed by that query.”

» Apparently Google's documentation explicitly recommends not to use `LIMIT` as a cost control mechanism for non-clustered tables. Secondly, BigQuery has a minimum billing amount of 10MB for

on-demand queries.

» For the `EXTERNAL TABLE`, BigQuery is querying the underlying external data rather than Google's own optimized native storage. Conceptually, this means that the `.csv` file must be read and parsed to produce the tabular data required by the query, rather than being accessed through BigQuery's native columnar storage.

Exercici 5: Adaptació de Sintaxi "Reporting"

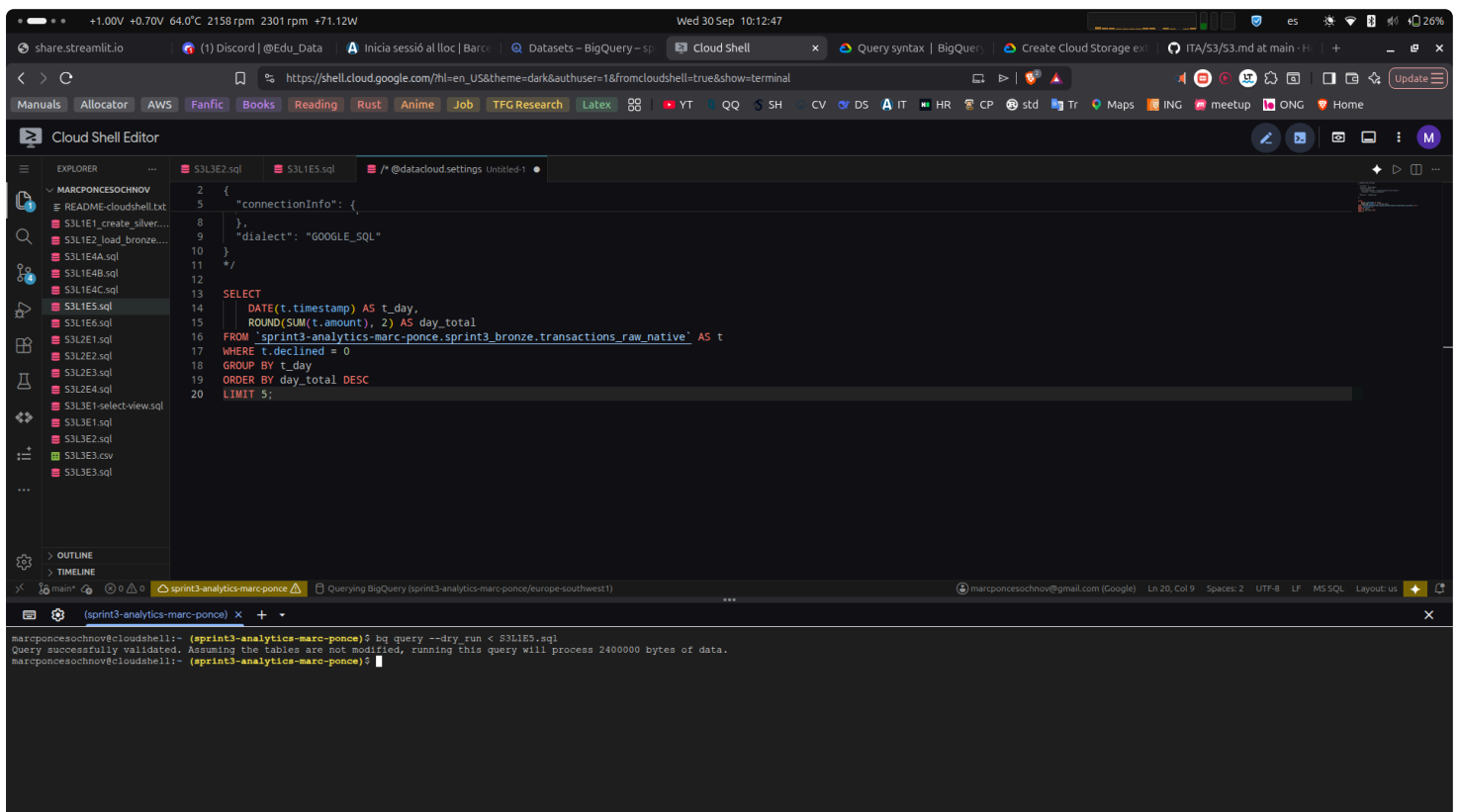
El teu cap vol saber quins van ser els **5 dies amb més ingressos de l'any 2021**.

🦾 Repte:

Probablement, el camp `timestamp` és un `STRING`. Hauràs d'investigar funcions de BigQuery (`SUBSTR`, `CAST`, `PARSE_TIMESTAMP`) per filtrar l'any i agrupar per data correctament.

» The `timestamp` field is, in fact, not a `string`. BigQuery had loaded it into the proper `TIMESTAMP` form.

```
SELECT
    DATE(t.timestamp) AS t_day,
    ROUND(SUM(t.amount), 2) AS day_total
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
WHERE t.declined = 0
GROUP BY t_day
ORDER BY day_total DESC
LIMIT 5;
```



The screenshot shows a Cloud Shell terminal window with a dark theme. The top bar displays system information like voltage, temperature, and CPU/RPM. The browser tabs at the top include 'share.streamlit.io', 'Discord', 'Edu_Data', 'Inicia sessió al lloc | Barce...', 'Datasets - BigQuery - sp...', 'Cloud Shell', 'Query syntax | BigQuer...', 'Create Cloud Storage ext...', and 'ITA/S3/S3.md at main · H...'. The address bar shows the Cloud Shell URL. The main editor area contains a SQL query in a file named 'S3L1E5.sql'. The query is the same as the one shown in the previous block. The terminal output at the bottom shows the command to run the query and the successful result.

```
marcponcesochnov@cloudshell:~$ bq query --dry_run < S3L1E5.sql
Query successfully validated. Assuming the tables are not modified, running this query will process 2400000 bytes of data.
marcponcesochnov@cloudshell:~$
```

The screenshot shows a Google Cloud Shell Editor window. The left sidebar contains a file explorer with various SQL files. The main editor area displays a SQL query in a dark theme. The query is as follows:

```

2 {
3   "connectionInfo": {
4     "dialect": "GOOGLE_SQL"
5   },
6   "query": "
7     SELECT
8       DATE(t.timestamp) AS t_day,
9       ROUND(SUM(t.amount), 2) AS day_total
10    FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
11   WHERE t.declined = 0
12   GROUP BY t_day
13   ORDER BY day_total DESC
14   LIMIT 5;
15 "
16 }

```

Below the query editor, the 'QUERY RESULTS' tab is active, showing a table with 5 rows and 2 columns: 't_day' and 'day_total'.

Row	t_day	day_total
1	2022-12-13	14337.44
2	2019-11-18	13591.32
3	2023-02-20	13332.59
4	2017-12-20	13318.43
5	2019-03-18	12680.95

» It took me a few tries to execute the right code (I was correcting), and it seems that given that I was querying pretty much the same information, it was cached somewhere between the database and my endpoint, thus I was not billed for this last query.

“../Pasted image 20260930101433.png” could not be found.

Exercici 6: Consultes Complexes

Necessitem un informe que creui dades.

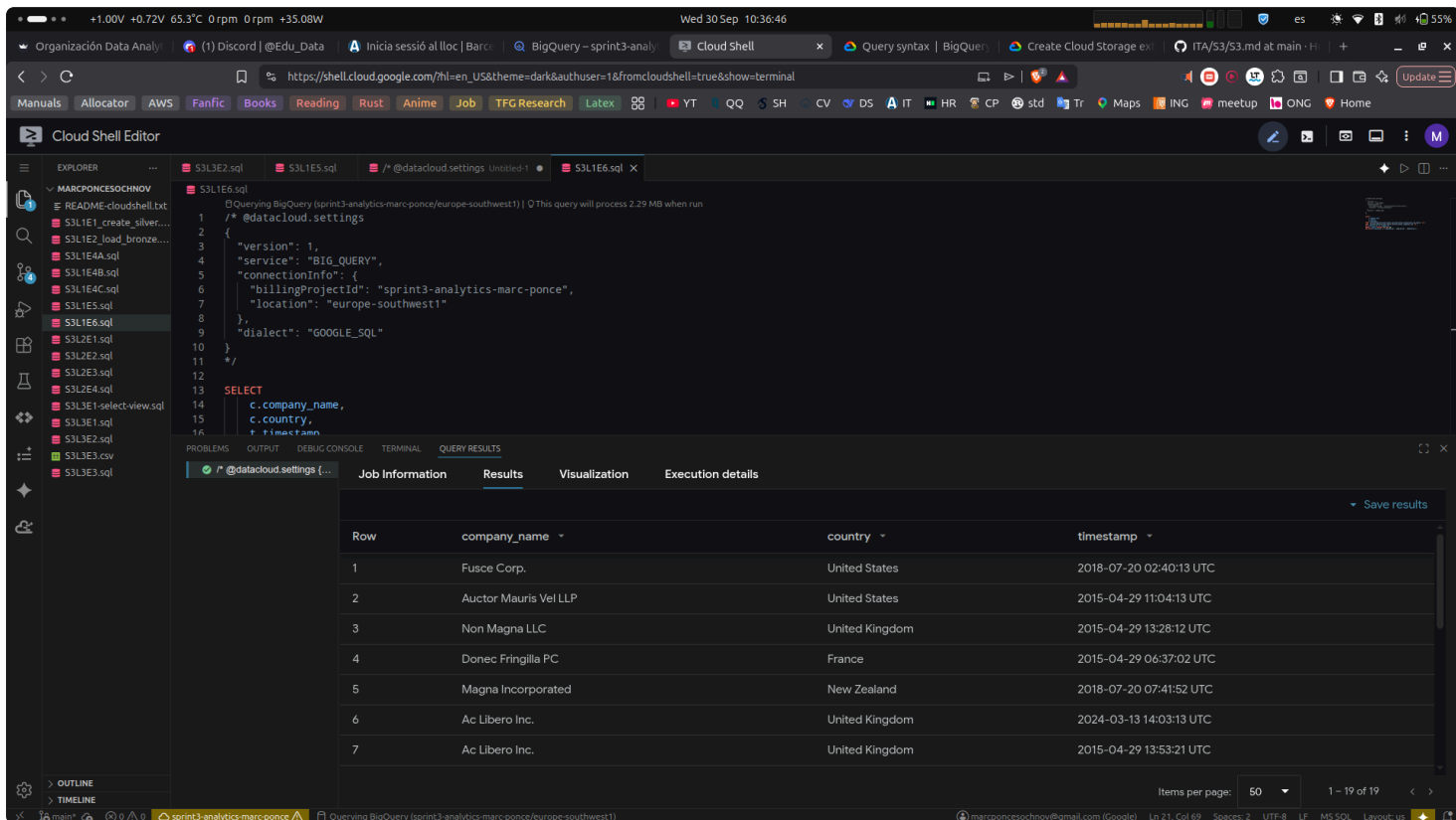
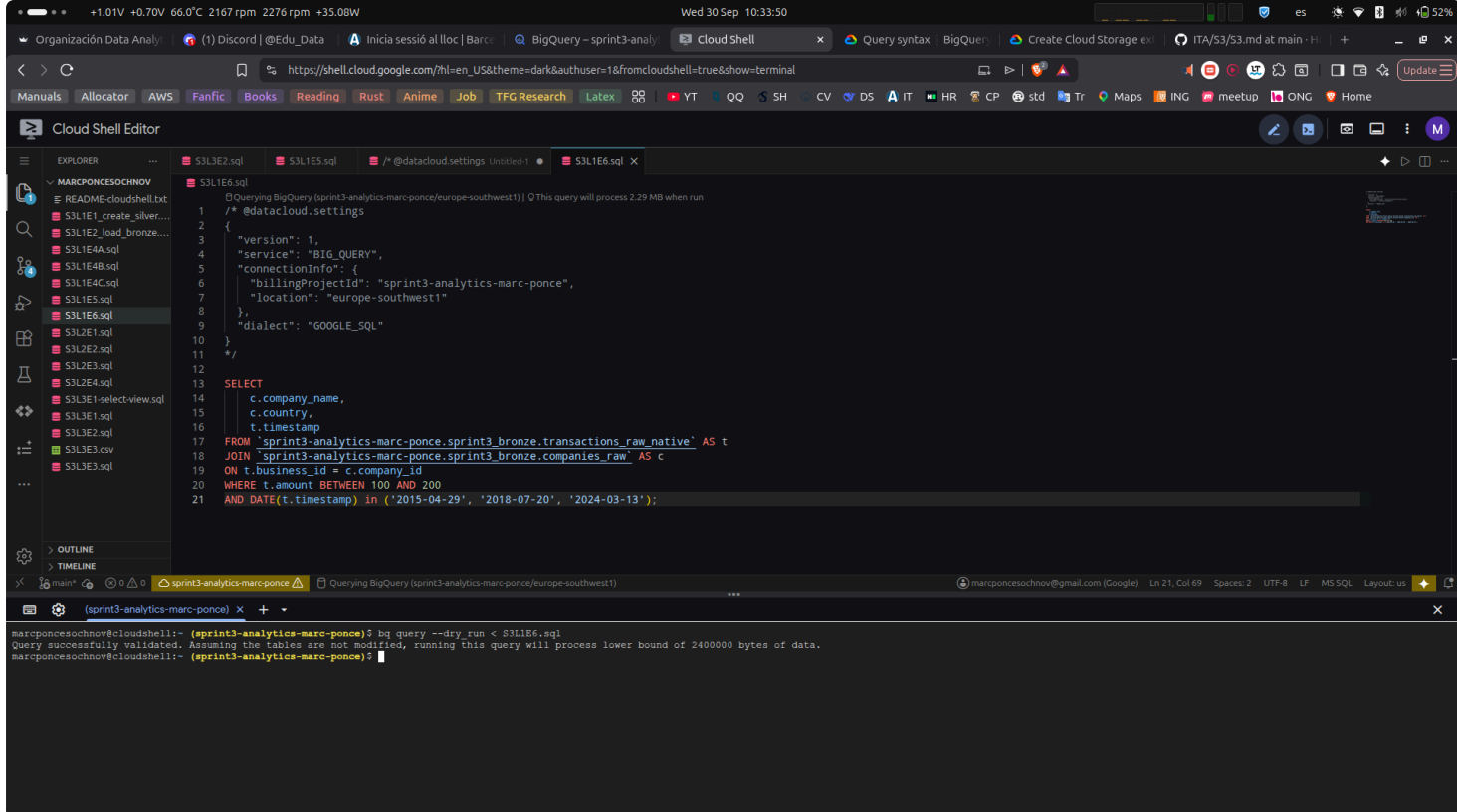
Tasca:

Llista el nom, país i data de les transaccions realitzades per empreses que van fer operacions entre 100 i 200 euros en alguna d'aquestes dates: 29-04-2015, 20-07-2018 o 13-03-2024.

```

SELECT
  c.company_name,
  c.country,
  t.timestamp
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c
ON t.business_id = c.company_id
WHERE t.amount BETWEEN 100 AND 200
AND DATE(t.timestamp) in ('2015-04-29', '2018-07-20', '2024-03-13');

```



The screenshot shows the Google Cloud Shell interface. The left sidebar contains a file explorer with various SQL files. The main editor displays a BigQuery query in a file named `S3L1E6.sql`. The query is a SELECT statement that joins `sprint3_analytics_marc_ponce.europe_southwest1.sprint3_bronze.transactions_raw_native` with `sprint3_analytics_marc_ponce.sprint3_bronze.companies_raw` and `sprint3_analytics_marc_ponce.europe_southwest1.european_users_raw`. The query filters for transactions with an amount between 100 and 200 and a timestamp in the years 2015, 2018, or 2024. The query is executed, and the bottom panel shows the job information.

```

1  /* @datacloud.settings
2  {
3    "version": 1,
4    "service": "BIG_QUERY",
5    "connectionInfo": {
6      "billingProjectId": "sprint3-analytics-marc-ponce",
7      "location": "europe-southwest1"
8    },
9    "dialect": "GOOGLE_SQL"
10 }
11 */
12
13 SELECT
14   c.company_name,
15   c.country,
16   t.timestamp

```

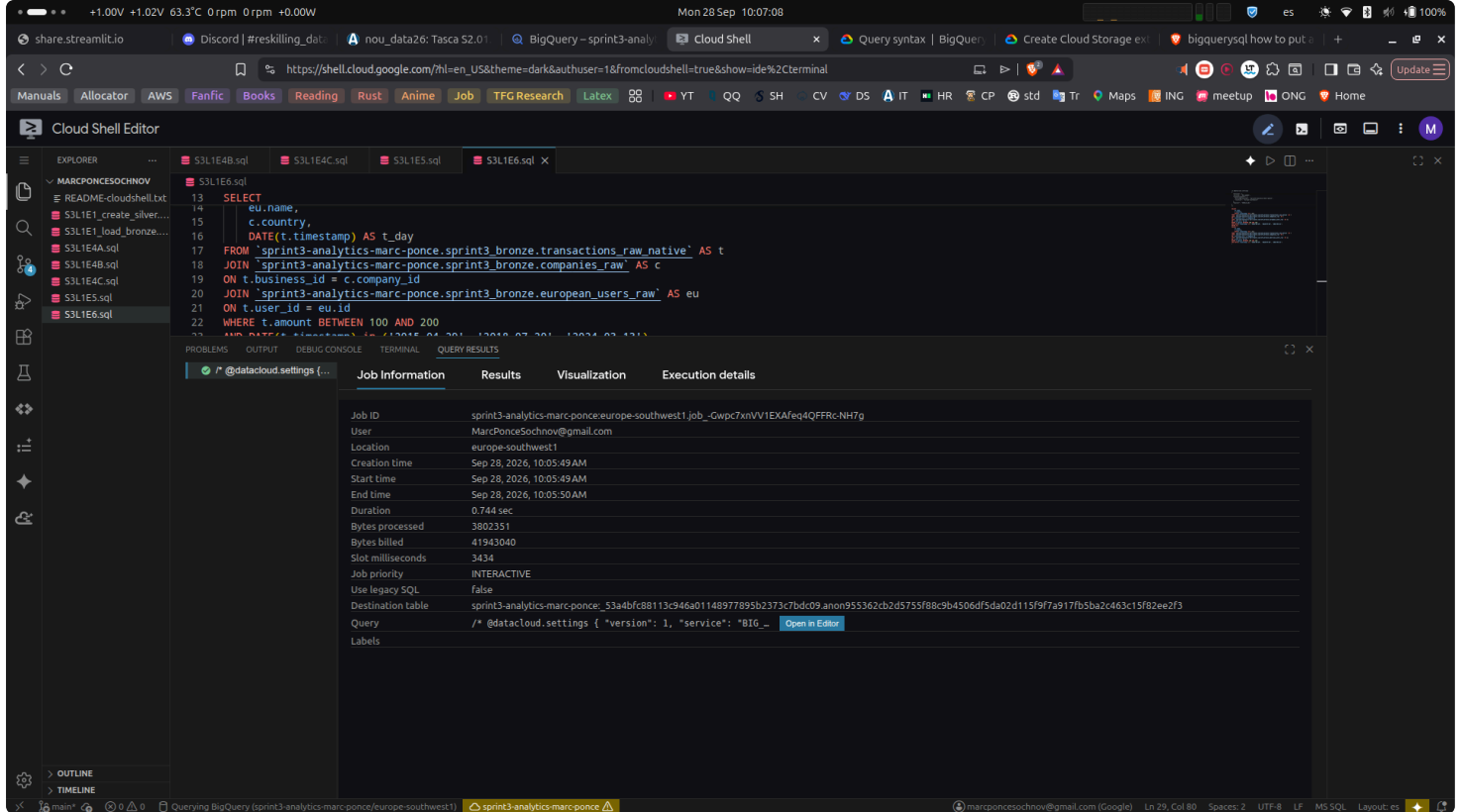
Job ID	sprint3-analytics-marc-ponce:europe-southwest1 Job_RNxA6cUXSVP-Qqpl0A1r2yjsBBI
User	MarcPonceSochnov@gmail.com
Location	europe-southwest1
Creation time	Sep 30, 2026, 10:35:36AM
Start time	Sep 30, 2026, 10:35:36AM
End time	Sep 30, 2026, 10:35:37AM
Duration	0.459 sec
Bytes processed	2410338
Bytes billed	20971520
Slot milliseconds	455
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce_53a4bfc88113c946a01148977895b2373c7bdc09.anon05213dceac4f1cf9ec5ad626e2adb46d270a8eeb9986f51248632d68a14b61bc
Query	/* @datacloud.settings { "version": 1, "service": "BIG_...
Labels	

» This is the old solution: for some reason when asked about the names, I thought of the names of the users not the name of the companies.

```

SELECT
  eu.name,
  c.country,
  DATE(t.timestamp) AS t_day
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c
ON t.business_id = c.company_id
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw` AS eu
ON t.user_id = eu.id
WHERE t.amount BETWEEN 100 AND 200
AND DATE(t.timestamp) in ('2015-04-29', '2018-07-20', '2024-03-13')
UNION ALL
SELECT
  au.name,
  c.country,
  DATE(t.timestamp) AS t_day
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c
ON t.business_id = c.company_id
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.american_users_raw` AS us
ON t.user_id = au.id
WHERE t.amount BETWEEN 100 AND 200
AND DATE(t.timestamp) in ('2015-04-29', '2018-07-20', '2024-03-13');

```

» It has occurred me that the old version code can be improved:

```
WITH
  tc AS (
    SELECT
      t.user_id,
      c.country,
      t.timestamp
    FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native`
  AS t
    JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c
    ON t.business_id = c.company_id
    WHERE t.amount BETWEEN 100 AND 200
    AND DATE(t.timestamp) IN ('2015-04-29', '2018-07-20', '2024-03-13')
  )
SELECT
  eu.name,
  tc.country,
  tc.timestamp
FROM tc
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw` AS eu
ON tc.user_id = eu.id
UNION ALL
SELECT
  us.name,
  tc.country,
  tc.timestamp
FROM tc
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.american_users_raw` AS us
ON tc.user_id = us.id;
```

Nivell 2: Neteja i Transformació (ELT)

Escenari:

Les dades brutes tenen problemes de qualitat (símbols de moneda, usuaris duplicats, dates incorrectes). Actua com a Data Engineer per netejar-ho tot.

Exercici 1: Neteja de Productes (Data Quality)

Crearem la capa neta ("Silver") per als productes. Tens la taula `sprint3_bronze.products_raw` carregada des del CSV. L'equip de Data Governance et demana crear una taula de productes neta a `sprint3_silver.products_clean` que compleixi aquestes regles de qualitat:

- **Estandardització de Noms:** La columna original `id` és ambigua; reanomena-la a `product_id`. La columna `product_name` simplifica-la a `name`.
- **Neteja d'IDs:** El camp `warehouse_id` té el format antic "WH-4". Elimina el prefix "WH-" i converteix el valor a nombre enter (INT64).
- **Garantia de Preu:** Assegura't que `price` és un número `FLOAT64`, sense símbols de moneda.
- **Altres columnes:** Conserva el camp `weight` (pes) tal qual.

```
CREATE OR REPLACE TABLE
    `sprint3-analytics-marc-ponce.sprint3_silver.products_clean`
AS SELECT
    p.id AS product_id,
    p.product_name AS name,
    CAST(REGEXP_REPLACE(p.warehouse_id, r'^0-9]', '') AS INT64) AS warehouse_id,
    CAST(price AS FLOAT64) AS price,
    p.colour,
    p.category,
    p.weight,
    p.brand,
    p.cost,
    p.launch_date
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.products_raw` AS p;
```

Mon 28 Sep 10:34:07

share.streamlit.io | Discord | #reskilling_data | nou_data26: Tasca S2.01 | BigQuery - sprint3-analy | cloud Shell | Query syntax | BigQuery | Create Cloud Storage ext | regex pattern to remove | Update

Manuals | Allocator | AWS | Fanfic | Books | Reading | Rust | Anime | Job | TFG Research | Latex | YT | QQ | SH | CV | DS | IT | HR | CP | std | Tr | Maps | ING | meetup | ONG | Home

Cloud Shell Editor

EXPLORER

MARCPONCESOCHNOV

README-cloudshell.txt

S3L1E1_create_silver...

S3L1E2_load_bronze...

S3L1E4A.sql

S3L1E4B.sql

S3L1E4C.sql

S3L1E5.sql

S3L1E6.sql

S3L2E1.sql

OUTLINE

TIMELINE

S3L2E1.sql

Querying BigQuery (sprint3-analytics-marc-ponce/europe-southwest1) | This query will process 8.88 KB when run

1 /* @datacloud.settings

2 {

3 "version": 1,

4 "service": "BIG_QUERY",

5 "connectionInfo": {

6 "billingProjectId": "sprint3-analytics-marc-ponce",

7 "location": "europe-southwest1"

8 },

9 "dialect": "GOOGLE_SQL"

10 }

11 */

12

13

14 CREATE OR REPLACE TABLE

15 'sprint3-analytics-marc-ponce.sprint3_silver.products_clean'

16 AS SELECT

17 p.id AS product_id,

18 p.product_name AS name,

19 CAST(REGEXP_REPLACE(p.warehouse_id, r'[^0-9]', '') AS INT64) AS warehouse_id,

20 CAST(price AS FLOAT64) AS price,

21 p.colour,

22 p.category,

23 p.weight,

24 p.brand,

25 p.cost,

26 p.launch_date

FROM 'sprint3-analytics-marc-ponce.sprint3_bronze.products_raw' AS p;

main | sprint3-analytics-marc-ponce | Querying BigQuery (sprint3-analytics-marc-ponce/europe-southwest1) | marcponcesochnov@gmail.com (Google) | Ln 24, Col 12 | Spaces: 4 | UTF-8 | LF | MS SQL | Layout: es

marcponcesochnov@cloudshell:~\$ (sprint3-analytics-marc-ponce)\$ bq query --location=europe-southwest1 --dry_run < S3L1E6.sql

marcponcesochnov@cloudshell:~\$ (sprint3-analytics-marc-ponce)\$ bq query --location=europe-southwest1 --dry_run < S3L2E1.sql

Query successfully validated. Assuming the tables are not modified, running this query will process 9095 bytes of data.

marcponcesochnov@cloudshell:~\$ (sprint3-analytics-marc-ponce)\$

Mon 28 Sep 10:36:21

share.streamlit.io | Discord | #reskilling_data | nou_data26: Tasca S2.01 | BigQuery - sprint3-analy | cloud Shell | Query syntax | BigQuery | Create Cloud Storage ext | regex pattern to remove | Update

Manuals | Allocator | AWS | Fanfic | Books | Reading | Rust | Anime | Job | TFG Research | Latex | YT | QQ | SH | CV | DS | IT | HR | CP | std | Tr | Maps | ING | meetup | ONG | Home

Cloud Shell Editor

EXPLORER

MARCPONCESOCHNOV

README-cloudshell.txt

S3L1E1_create_silver...

S3L1E2_load_bronze...

S3L1E4A.sql

S3L1E4B.sql

S3L1E4C.sql

S3L1E5.sql

S3L1E6.sql

S3L2E1.sql

OUTLINE

TIMELINE

S3L2E1.sql

Querying BigQuery (sprint3-analytics-marc-ponce/europe-southwest1) | This query will process 8.88 KB when run

1 /* @datacloud.settings

2 {

3 "version": 1,

4 "service": "BIG_QUERY",

5 "connectionInfo": {

6 "billingProjectId": "sprint3-analytics-marc-ponce",

7 "location": "europe-southwest1"

8 },

9 "dialect": "GOOGLE_SQL"

10 }

11 */

12

13

14 CREATE OR REPLACE TABLE

15 'sprint3-analytics-marc-ponce.sprint3_silver.products_clean'

16 AS SELECT

17 p.id AS product_id,

18 p.product_name AS name,

19 CAST(REGEXP_REPLACE(p.warehouse_id, r'[^0-9]', '') AS INT64) AS warehouse_id,

20 CAST(price AS FLOAT64) AS price,

21 p.colour,

22 p.category,

23 p.weight,

24 p.brand,

25 p.cost,

26 p.launch_date

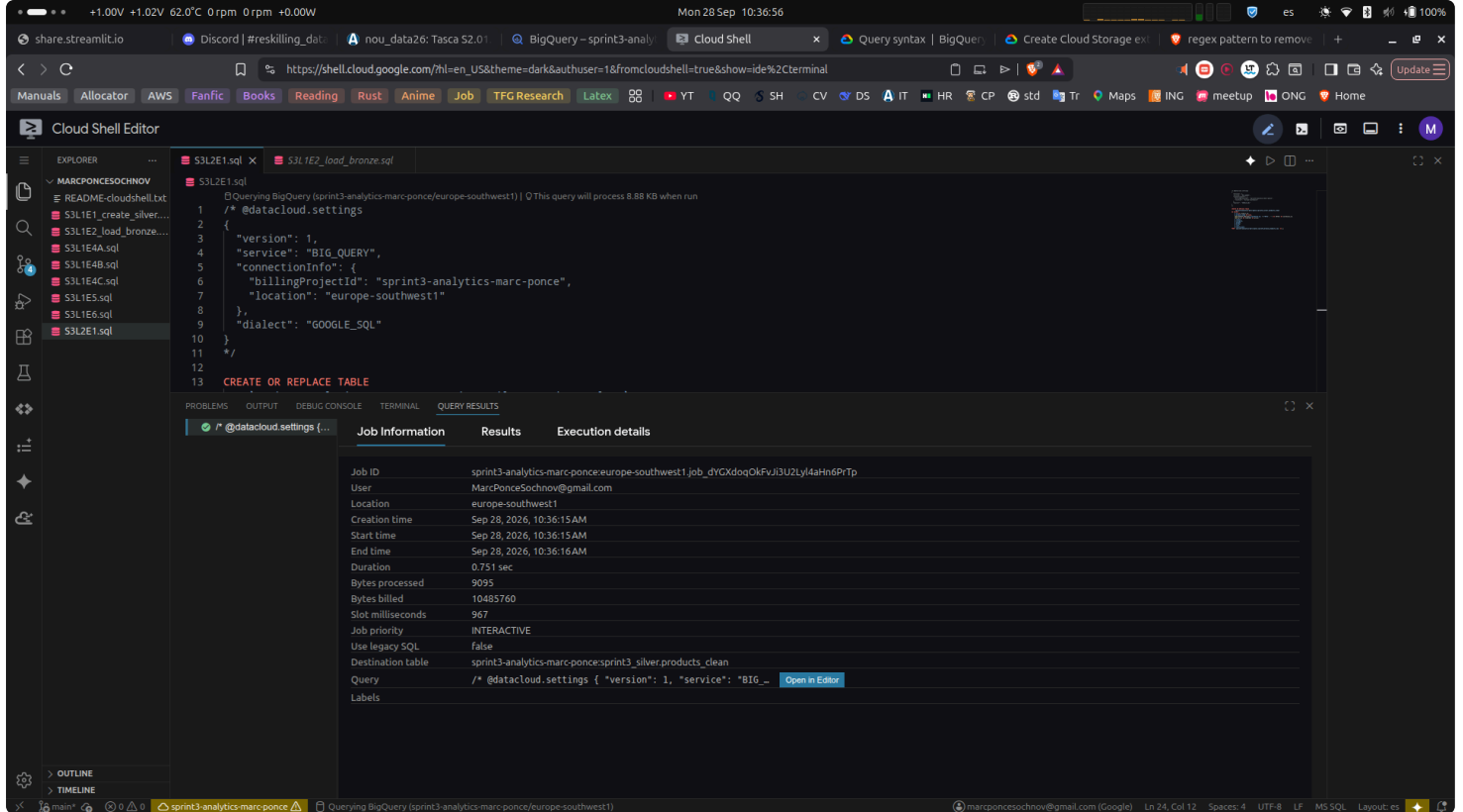
FROM 'sprint3-analytics-marc-ponce.sprint3_bronze.products_raw' AS p;

main | sprint3-analytics-marc-ponce | Querying BigQuery (sprint3-analytics-marc-ponce/europe-southwest1) | marcponcesochnov@gmail.com (Google) | Ln 24, Col 12 | Spaces: 4 | UTF-8 | LF | MS SQL | Layout: es

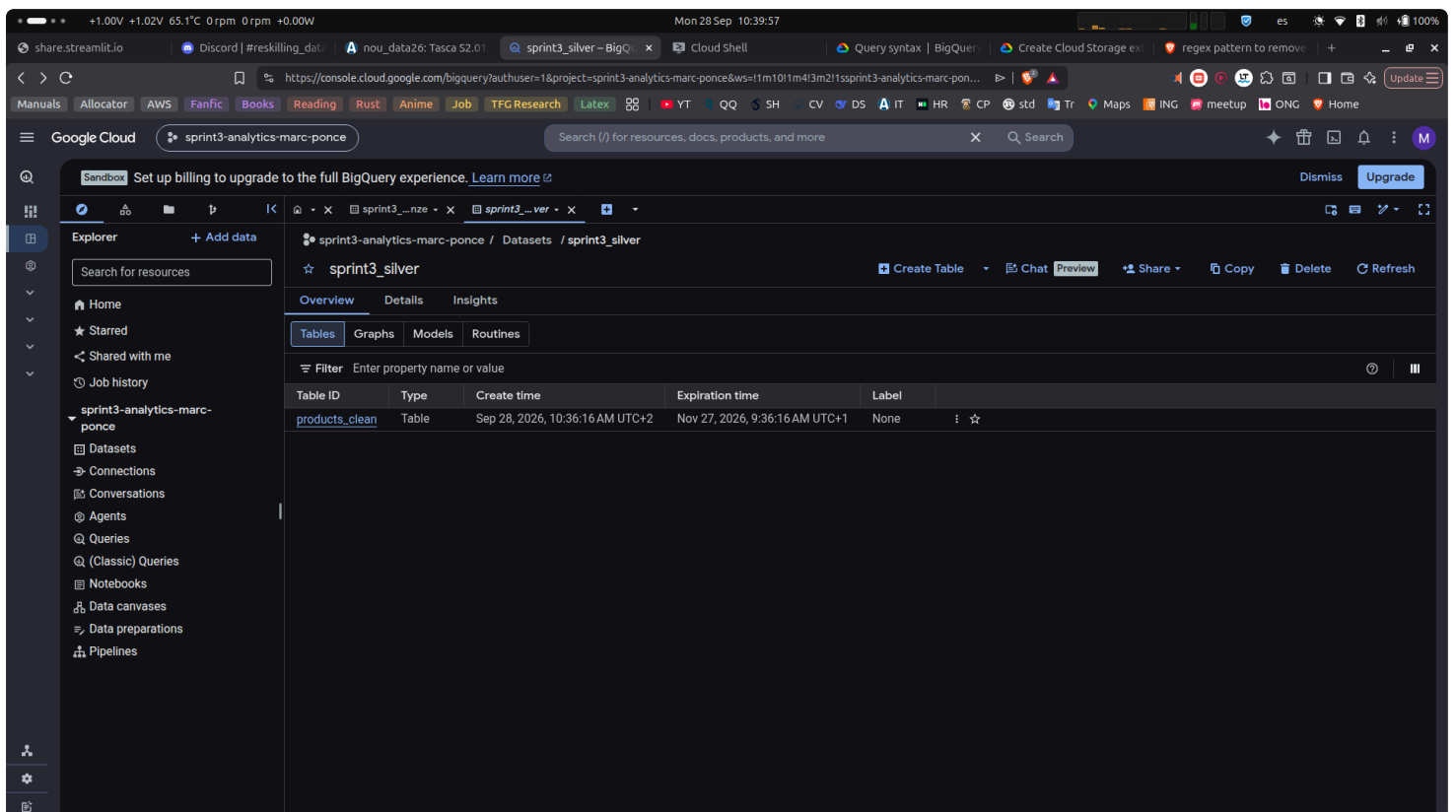
PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | QUERY RESULTS

Job Information | Results | Execution details

This statement created a new table named products_clean.



» Last, we check that the table is where it should be:



Exercici 2: Creació de Transaccions Netes (Capa Silver)

Escenari:

La taula `transactions_raw` (Bronze) és insegura: l'import (`amount`) podria tenir lletres, les coordenades podrien fallar i la data és un text difícil de filtrar. Crearem la taula definitiva `sprint3_silver.transactions_clean` que garanteixi el següent.

- **Estandardització de Noms:** La columna original `id` és ambigua; reanomena-la a `transaction_id`.

- **Robustesa en Imports:** Utilitza `SAFE_CAST` al camp `amount`. Si falla, substitueix-lo per 0 (usant `IFNULL`).
- **Dates Reals:** Converteix el camp `timestamp` (que és `STRING`) a tipus `TIMESTAMP` real.
- **Coordenades:** Assegura que `lat` i `longitude` siguin `FLOAT64` (utilitza `SAFE_CAST` per seguretat).
- **Desglossa els Productes:** Transforma la cadena de text `product_ids` en un `ARRAY` d'enters.
- *Input:* "1, 2, 3" `STRING`
- *Output:* [1, 2, 3] `ARRAY<INT>`
- **Resta de camps:** Mantén-los igual.

```
CREATE OR REPLACE TABLE
```

```
  `sprint3-analytics-marc-ponce.sprint3_silver.transactions_clean`
```

```
AS SELECT
```

```
  t.id AS transaction_id,
```

```
  t.business_id,
```

```
  t.card_id,
```

```
  timestamp,
```

```
  IFNULL(SAFE_CAST(t.amount AS FLOAT64), 0) AS amount,
```

```
  t.declined,
```

```
  ARRAY(
```

```
    SELECT SAFE_CAST(TRIM(product_id) AS INT64)
```

```
    FROM UNNEST(SPLIT(t.product_ids, ',')) AS product_id
```

```
  ) AS product_ids,
```

```
  user_id,
```

```
  SAFE_CAST(t.lat AS FLOAT64) AS lat,
```

```
  SAFE_CAST(t.longitude AS FLOAT64) AS longitude
```

```
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t;
```

» `SPLIT(t.product_ids, ',')` separates the IDs and transforms the field into an `ARRAY<STRING>` . Unfortunately, BigQuery's `SQL` has no function such as `MAP(...)` . Therefore, to cast each `product_id` into an `INT` , we have to transform the field into a virtual table with `UNNEST(...)` , which transforms each element of the array into a row of the table. The cast within a `SELECT` query is then straightforward—which leads to the last operation, which uses `ARRAY(...)` to transform the virtual

table back into an ARRAY<INT64> data type.

The screenshot shows the Cloud Shell Editor interface. The Explorer pane on the left lists files: README-cloudshell.txt, S3L1E1_create_silver..., S3L1E2_load_bronze..., S3L1E4A.sql, S3L1E4B.sql, S3L1E4C.sql, S3L1E5.sql, S3L1E6.sql, S3L2E1.sql, and S3L2E2.sql. The main editor displays the content of S3L2E2.sql, which includes a BigQuery settings block and a SQL query to create a table and insert data. The status bar at the bottom indicates the query is running on the 'sprint3-analytics-marc-ponce' project in the 'europe-southwest1' region.

```
1 /* @datacloud.settings
2 {
3   "version": 1,
4   "service": "BIG_QUERY",
5   "connectionInfo": {
6     "billingProjectId": "sprint3-analytics-marc-ponce",
7     "location": "europe-southwest1"
8   },
9   "dialect": "GOOGLE_SQL"
10 }
11 */
12
13 CREATE OR REPLACE TABLE
14 `sprint3-analytics-marc-ponce.sprint3_silver.transactions_clean`
15 AS
16 SELECT
17   t.id AS transaction_id,
18   t.business_id,
19   t.card_id,
20   timestamp,
21   IFNULL(SAFE_CAST(t.amount AS FLOAT64), 0) AS amount,
22   t.declined,
23   ARRAY(
24     SELECT SAFE_CAST(TRIM(product_id) AS INT64)
25     FROM UNNEST(SPLIT(t.product_ids, ',')) AS product_id
26   ) AS product_ids,
27   user_id,
28   SAFE_CAST(t.lat AS FLOAT64) AS lat,
29   SAFE_CAST(t.longitude AS FLOAT64) AS longitude
```

marcponcesochnov@cloudshell:~\$ bq query --location=europe-southwest1 --dry run < S3L2E2.sql
Query successfully validated. Assuming the tables are not modified, running this query will process 11393306 bytes of data.
marcponcesochnov@cloudshell:~\$

This screenshot shows the same Cloud Shell Editor interface, but with the 'QUERY RESULTS' pane open at the bottom. The pane displays a message indicating that the statement successfully created a new table named 'products_clean'. The status bar at the bottom shows the query has completed.

Job Information Results Execution details

This statement created a new table named products_clean.

Cloud Shell Editor

```

1  /* @datacloud.settings
2  3  {
3  4    "version": 1,
4  5    "service": "BIG_QUERY",
5  6    "connectionInfo": {
6  7      "billingProjectId": "sprint3-analytics-marc-ponce",
7  8      "location": "europe-southwest1"
8  9    },
9  10   "dialect": "GOOGLE_SQL"
10  11 }
11  12 */
12  13 CREATE OR REPLACE TABLE

```

Job Information

Job ID	sprint3-analytics-marc-ponce:europe-southwest1.job_dYGXdoqOKFvJ3U2Lyl4aHn6PrTp
User	MarcPonceSochnov@gmail.com
Location	europe-southwest1
Creation time	Sep 28, 2026, 10:36:15AM
Start time	Sep 28, 2026, 10:36:15AM
End time	Sep 28, 2026, 10:36:16AM
Duration	0.751 sec
Bytes processed	9095
Bytes billed	10485760
Slot milliseconds	967
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce:sprint3_silver.products_clean
Query	/* @datacloud.settings { "version": 1, "service": "BIG_...
Labels	

» Finally we check that the table is in place:

Google Cloud

sprint3-analytics-marc-ponce

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sprint3-analytics-marc-ponce

Datasets

Connections

Conversations

Agents

Queries

(Classic) Queries

Notebooks

Data canvases

Data preparations

Pipelines

sprint3-analytics-marc-ponce / Datasets / sprint3_silver / Tables / transactions_clean

transactions_clean

Schema Details Preview Insights Lineage Data Profile Data Quality

Filter Enter property name or value

Field name	Type	Mode	Description	Key	Collation	Default Value	Data Governance Tags	Policy Tags	Data Policies
transaction_id	STRING	NULLABLE	-	-	-	-	-	-	-
business_id	STRING	NULLABLE	-	-	-	-	-	-	-
card_id	STRING	NULLABLE	-	-	-	-	-	-	-
timestamp	TIMESTAMP	NULLABLE	-	-	-	-	-	-	-
amount	FLOAT	NULLABLE	-	-	-	-	-	-	-
declined	INTEGER	NULLABLE	-	-	-	-	-	-	-
product_ids	INTEGER	REPEATED	-	-	-	-	-	-	-
user_id	INTEGER	NULLABLE	-	-	-	-	-	-	-
lat	FLOAT	NULLABLE	-	-	-	-	-	-	-
longitude	FLOAT	NULLABLE	-	-	-	-	-	-	-

Edit schema View row access policies Describe data

» REPEATED means that the field holds zero or more values of the specified type (i.e., INT64 in this example).

Exercici 3: Unificació d'Usuaris (UNION)

Tenim els usuaris fragmentats per regió.

Tasca:

Crea la taula `sprint3_silver.users_combined`. Utilitza `UNION ALL` per unificar els usuaris dels EUA i Europa en una única llista mestra. Afegeix una columna calculada `origin` per saber d'on venen.

- **Estandardització de Noms:** La columna original `id` és ambigua; reanomena-la a `user_id`.

```
CREATE OR REPLACE TABLE
`sprint3-analytics-marc-ponce.sprint3_silver.users_combined`
AS SELECT
  eu.id AS user_id,
  eu.name,
  eu.surname,
  eu.phone,
  eu.email,
  eu.birth_date,
  eu.country,
  eu.city,
  eu.postal_code,
  eu.address,
  'EU' AS origin
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw` AS eu
UNION ALL
SELECT
  au.id AS user_id,
  au.name,
  au.surname,
  au.phone,
  au.email,
  au.birth_date,
  au.country,
  au.city,
  au.postal_code,
  au.address,
  'US' AS origin
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.american_users_raw` AS au;
```

```
share.streamlit.io | Discord | #reskilling_data_an | Inicia sessió al lloc | Barcelona | BigQuery - sprint3-analytics | Cloud Shell | Query syntax | BigQuery | Create Cloud Storage extern | + | Update
```

Cloud Shell Editor

EXPLORER

- MARCPONCESOCHNOV
 - README-cloudshell.txt
 - S3L1E1_create_silver...
 - S3L1E2_load_bronze...
 - S3L1E4A.sql
 - S3L1E4B.sql
 - S3L1E4C.sql
 - S3L1E5.sql
 - S3L1E6.sql
 - S3L2E1.sql
 - S3L2E2.sql
 - S3L2E3.sql
 - S3L2E4.sql
 - S3L3E1-select-view.sql
 - S3L3E1.sql
 - S3L3E2.sql
 - S3L3E3.csv
 - S3L3E3.sql

S3L2E3.sql

```
10 }
11 */
12
13 CREATE OR REPLACE TABLE
14 'sprint3-analytics-marc-ponce.sprint3_silver.users_combined'
15 AS
16 SELECT
17   eu.id AS user_id,
18   eu.name,
19   eu.surname,
20   eu.phone,
21   eu.email,
22   eu.birth_date,
23   eu.country,
24   eu.city,
25   eu.postal_code,
26   eu.address,
27   'EU' AS origin
28 FROM 'sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw' AS eu
29 UNION ALL
30 SELECT
31   au.id AS user_id,
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL QUERY RESULTS

main* 0 0 0 sprint3-analytics-marc-ponce Querying BigQuery (sprint3-analytics-marc-ponce/europe-southwest1) marcponcesochnov@gmail.com (Google) Ln 39, Col 16 Spaces: 2 UTF-8 LF MS SQL Layout: us

marcponcesochnov@cloudshell:~\$ (sprint3-analytics-marc-ponce) bq query --dry-run < S3L2E3.sql
Query successfully validated. Assuming the tables are not modified, running this query will process lower bound of 0 bytes of data.
marcponcesochnov@cloudshell:~\$ (sprint3-analytics-marc-ponce)

```
share.streamlit.io | Discord | #reskilling_data_an | Inicia sessió al lloc | Barcelona | BigQuery - sprint3-analytics | Cloud Shell | Query syntax | BigQuery | Create Cloud Storage extern | + | Update
```

Cloud Shell Editor

EXPLORER

- MARCPONCESOCHNOV
 - README-cloudshell.txt
 - S3L1E1_create_silver...
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 - S3L1E4C.sql
 - S3L1E5.sql
 - S3L1E6.sql
 - S3L2E1.sql
 - S3L2E2.sql
 - S3L2E3.sql
 - S3L2E4.sql
 - S3L3E1-select-view.sql
 - S3L3E1.sql
 - S3L3E2.sql
 - S3L3E3.csv
 - S3L3E3.sql

S3L2E3.sql

```
10 }
11 */
12
13 CREATE OR REPLACE TABLE
14 'sprint3-analytics-marc-ponce.sprint3_silver.users_combined'
15 AS
16 SELECT
17   eu.id AS user_id,
18   eu.name,
19   eu.surname,
20   eu.phone,
21   eu.email,
22   eu.birth_date,
23   eu.country,
24   eu.city,
25   eu.postal_code,
26   eu.address,
27   'EU' AS origin
28 FROM 'sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw' AS eu
29 UNION ALL
30 SELECT
31   au.id AS user_id,
32   au.name,
33   au.surname,
34   au.phone,
35   au.email,
36   au.birth_date,
37   au.country,
38   au.city,
39   au.postal_code,
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL QUERY RESULTS

@datacloud.settings Job Information Results Execution details

This statement replaced the table named users_combined.

main* 0 0 0 sprint3-analytics-marc-ponce Querying BigQuery (sprint3-analytics-marc-ponce/europe-southwest1) marcponcesochnov@gmail.com (Google) Ln 39, Col 16 Spaces: 2 UTF-8 LF MS SQL Layout: us

Cloud Shell Editor interface showing a SQL query execution. The query is:

```
CREATE OR REPLACE TABLE
'sprint3-analytics-marc-ponce.sprint3_silver.users_combined'
AS SELECT
eu.id AS user_id,
eu.name,
eu.surname,
eu.phone,
eu.email,
eu.birth_date,
eu.country,
eu.city,
eu.postal_code,
```

The execution details panel shows the following information:

Job ID	sprint3-analytics-marc-ponce:europa-southwest1.job_1F3ukM9uKlbZvH4T1AOqDefmaXMy
User	MarcPonceSodhnov@gmail.com
Location	europa-southwest1
Creation time	Sep 29, 2026, 10:52:33 AM
Start time	Sep 29, 2026, 10:52:33 AM
End time	Sep 29, 2026, 10:52:35 AM
Duration	2.119 sec
Bytes processed	581675
Bytes billed	20971520
Slot milliseconds	1048
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-poncesprint3_silver.users_combined
Query	/* @datacloud.settings { "version": 1, "service": "BIG_...
Labels	

Google Cloud BigQuery console interface showing the schema of the table 'users_combined'.

Field name	Type	Mode	Description	Key	Collation	Default Value	Data Governance Tags	Policy Tags	Data Policies
user_id	INTEGER	NULLABLE	-	-	-	-	-	-	-
name	STRING	NULLABLE	-	-	-	-	-	-	-
surname	STRING	NULLABLE	-	-	-	-	-	-	-
phone	STRING	NULLABLE	-	-	-	-	-	-	-
email	STRING	NULLABLE	-	-	-	-	-	-	-
birth_date	STRING	NULLABLE	-	-	-	-	-	-	-
country	STRING	NULLABLE	-	-	-	-	-	-	-
city	STRING	NULLABLE	-	-	-	-	-	-	-
postal_code	STRING	NULLABLE	-	-	-	-	-	-	-
address	STRING	NULLABLE	-	-	-	-	-	-	-
origin	STRING	NULLABLE	-	-	-	-	-	-	-

Exercici 4: Materialització de Companyies i Targetes de Crèdit

Escenari:

Per completar el model de dades a la capa Silver, ens falten dues dimensions clau que actualment depenen de fitxers CSV externs: **Companyies** i **Targetes de Crèdit**. No podem dependre de fitxers solts per a l'anàlisi final. Hem d'importar aquestes dades a taules natives (Silver) i aprofitar per corregir formats.

Tasques:

-

- cal, reanomenar l' id segons correspongui.

The screenshot displays the Cloud Shell Editor interface. The Explorer pane on the left lists several SQL files, with `S3L2E4.sql` selected. The main editor area shows the content of `S3L2E4.sql`, which includes a BigQuery connection profile and two table creation statements.

```

1  /* @datacloud.settings
2  {
3    "version": 1,
4    "service": "BIG_QUERY",
5    "connectionInfo": {
6      "billingProjectId": "sprint3-analytics-marc-ponce",
7      "location": "europe-southwest1"
8    },
9    "dialect": "GOOGLE_SQL"
10  }
11  */
12
13  CREATE OR REPLACE TABLE
14    `sprint3-analytics-marc-ponce.sprint3_silver.companies_clean`
15  AS SELECT *
16  FROM `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c;
17
18  CREATE OR REPLACE TABLE
19    `sprint3-analytics-marc-ponce.sprint3_silver.credit_cards_clean`
20  AS SELECT *
21  FROM `sprint3-analytics-marc-ponce.sprint3_bronze.credit_cards_raw` AS c;

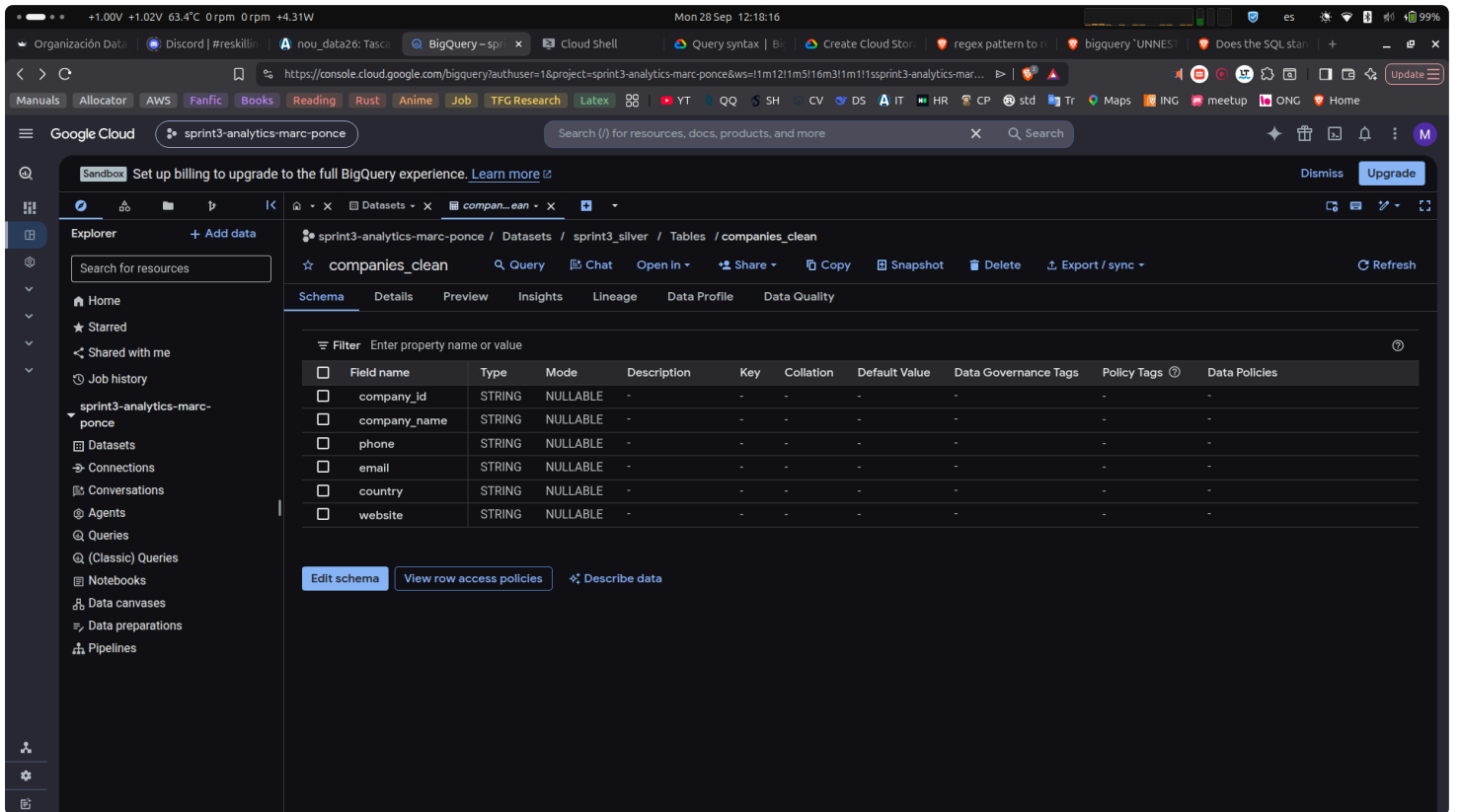
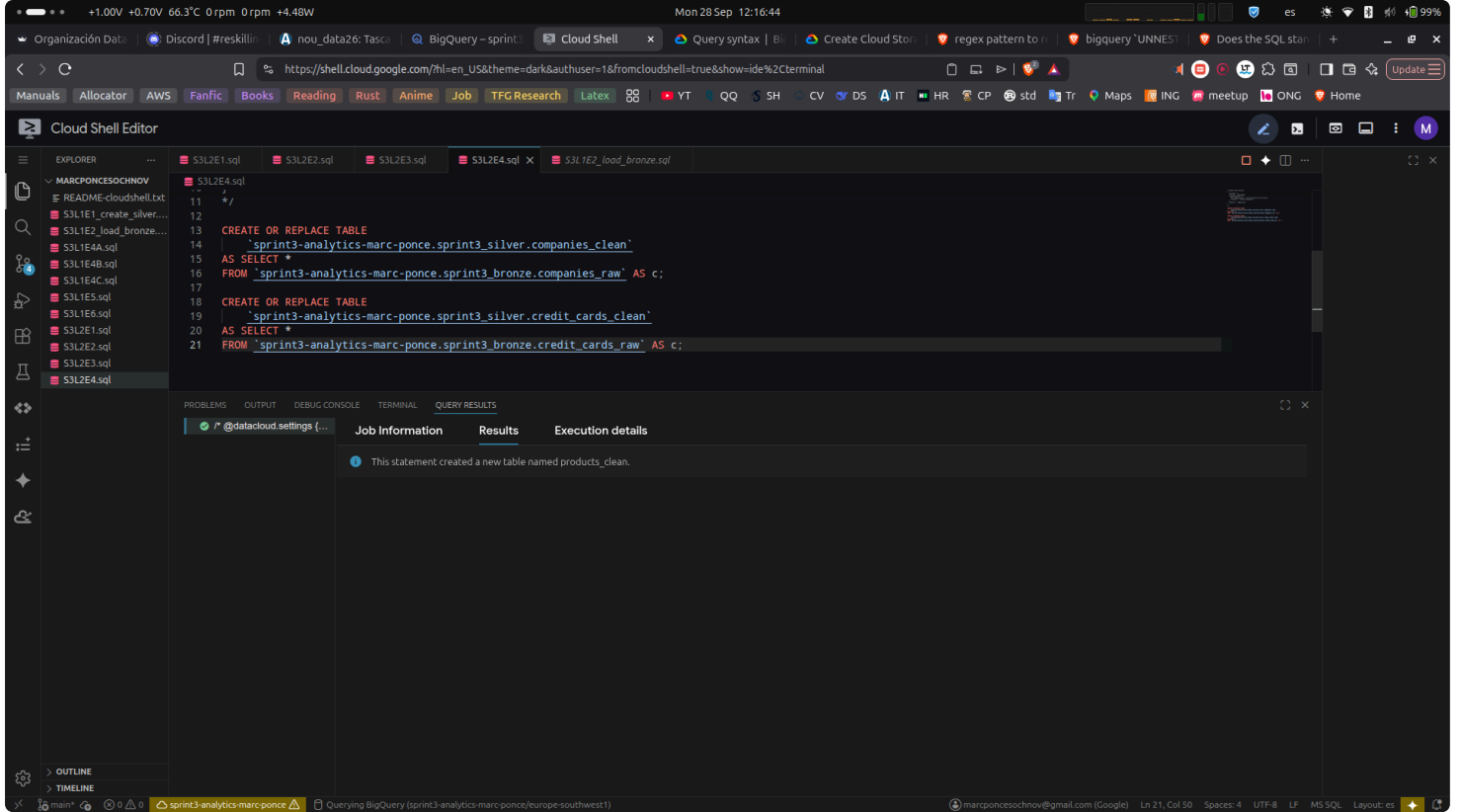
```

The terminal at the bottom shows the execution of the script:

```

rponce@cloudshell:~$ (sprint3-analytics-marc-ponce) bq query --location=europe-southwest1 --dry_run < S3L2E4.sql
Query successfully validated. Assuming the tables are not modified, running this query will process lower bound of 0 bytes of data.
rponce@cloudshell:~$ (sprint3-analytics-marc-ponce)

```



The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains navigation options like Explorer, Home, Starred, Shared with me, Job history, and various data management tools. The main area displays the schema for the 'credit_cards_clean' table. The schema table is as follows:

Field name	Type	Mode	Description	Key	Collation	Default Value	Data Governance Tags	Policy Tags	Data Policies
id	STRING	NULLABLE	-	-	-	-	-	-	-
user_id	INTEGER	NULLABLE	-	-	-	-	-	-	-
iban	STRING	NULLABLE	-	-	-	-	-	-	-
pan	STRING	NULLABLE	-	-	-	-	-	-	-
pin	INTEGER	NULLABLE	-	-	-	-	-	-	-
cvv	INTEGER	NULLABLE	-	-	-	-	-	-	-
track1	STRING	NULLABLE	-	-	-	-	-	-	-
track2	STRING	NULLABLE	-	-	-	-	-	-	-
expiring_date	DATE	NULLABLE	-	-	-	-	-	-	-

Nivell 3: Presentació de Dades i Creació de Vistes

Escenari:

Al nivell anterior vas actuar com a Data Engineer netejant les dades (*clean*). Ara toca posar-se el barret d'Analista de Negoci. El problema actual és que les taules estan separades i algunes tenen formats difícils d'explotar (com la llista de productes comprimida en una sola cel·la). En aquest nivell crearàs **Vistes** per facilitar l'accés a la informació i generaràs informes finals.

Objectius:

- Crear Vistes (`CREATE VIEW`) per encapsular lògica de negoci complexa.
- Resoldre el repte de dades agrupades en strings (`SPLIT` + `UNNEST`).
- Exportar resultats per a tercers (`Reverse ETL`).

Exercici 1: La Vista de Màrqueting (Lògica de Negoci)

Màrqueting necessita segmentar els clients corporatius, però no saben fer JOINS. La teva tasca és deixar-los la informació preparada.

Tasca:

Crea una vista anomenada `sprint3_gold.v_marketing_kpis` que mostri la següent informació per a cada companyia:

- **Nom de la companyia, Telèfon i País** (origen: `companies_clean`).
- **Mitjana de compra** (`AVG(amount)` de `transactions_clean`).
- **Classificació de Client (Lògica)**:
- Crea una columna calculada anomenada `client_tier` .
- Si la mitjana de compra és superior a **260€**, etiqueta com a **"Premium"**.
- Si és igual o inferior, etiqueta com a **"Standard"**.

👉 Entrega:

Realitza una consulta `SELECT *` sobre la teva nova vista, ordenant els resultats perquè apareguin primer els clients "Premium" i, dins d'aquests, els que tinguin major mitjana de compra.

```
CREATE OR REPLACE VIEW
`sprint3-analytics-marc-ponce.sprint3_gold.v_marketing_kpis`
AS WITH
a AS (
    SELECT
        c.company_id,
        c.company_name,
        c.phone,
        c.country,
        AVG(t.amount) AS avg_amount
    FROM `sprint3-analytics-marc-ponce.sprint3_silver.companies_clean` AS c
    JOIN `sprint3-analytics-marc-ponce.sprint3_silver.transactions_clean` AS t
    ON t.business_id = c.company_id
    WHERE t.declined = 0
    GROUP BY
        c.company_id,
        c.company_name,
        c.phone,
        c.country
)
SELECT
    a.company_name,
    a.phone,
    a.country,
    a.avg_amount,
    CASE
        WHEN a.avg_amount > 260 THEN 'Premium'
        ELSE 'Standard'
    END AS client_tier
FROM a;
```


Cloud Shell Editor interface showing a SQL query being executed. The query is:

```
CREATE OR REPLACE VIEW
`sprint3-analytics-marc-ponce.sprint3_gold.v_marketing_kpis`
AS WITH
  a AS (
    SELECT
```

The interface displays the query results in a table format, including job information and execution details.

```
SELECT *
FROM `sprint3-analytics-marc-ponce.sprint3_gold.v_marketing_kpis`
ORDER BY avg_amount DESC;
```

Cloud Shell Editor interface showing the execution results of the SQL query. The results are displayed in a table format, including job information and execution details.

Row	company_name	phone	country	avg_amount	client_tier
1	Ac Fermentum Incorporated	06 85 56 52 33	Germany	284.911333333333	Premium
2	Pretium Neque Corp.	07 77 48 55 28	Australia	275.5780637813211	Premium
3	Urna Conwallis Associates	06 01 24 77 04	United States	273.56692477876095	Premium
4	At Associates	09 56 61 10 65	New Zealand	272.73698481561865	Premium
5	Metus Vitae Associates	08 25 44 40 66	Australia	270.05187648456047	Premium
6	Aliquet Diam Limited	02 76 61 47 46	United States	269.2911447084236	Premium
7	Nec Luctus LLC	02 14 71 75 73	Norway	268.6048370927316	Premium
8	Neque Tellus Incorporated	04 43 18 34 19	Ireland	267.56334894613576	Premium
9	Cras Consulting	07 50 10 85 63	Belgium	267.3827966101694	Premium
10	Sed LLC	01 63 16 26 52	Belgium	266.6073509933775	Premium
11	Eget Ipsum Ltd	03 67 44 56 72	United States	266.2665454545458	Premium
12	Tortor Nunc Commodo Company	06 35 92 77 16	United States	264.2643749999997	Premium

Exercici 2: Rànquing de Productes (La Potència dels ARRAY s)

Escenari de Negoci:

L'equip de vendes vol optimitzar el catàleg. Necessiten un informe a la capa Gold que mostri el rendiment real de cada producte. Gràcies a la teva bona enginyeria a la capa Silver, la taula

transactions_clean ja té els IDs de producte perfectament organitzats en una llista (ARRAY). Ara toca explotar aquesta estructura.

Objectiu:

Crea la taula sprint3_gold.product_sales_ranking que contingui l'inventari complet de productes i quantes vegades s'ha venut cadascun.

Requisits de l'Informe:

- **Detall del Producte:** Ha d'incloure product_id , name , price i color (venen de la taula products_clean).
- **Mètrica de Negoci:** Una columna nova total_sold que compti quantes vegades apareix aquest producte a les transaccions.
 - **Integritat:** Han d'aparèixer tots els productes, fins i tot els que tenen 0 vendes (potser cal descatalogar-los).

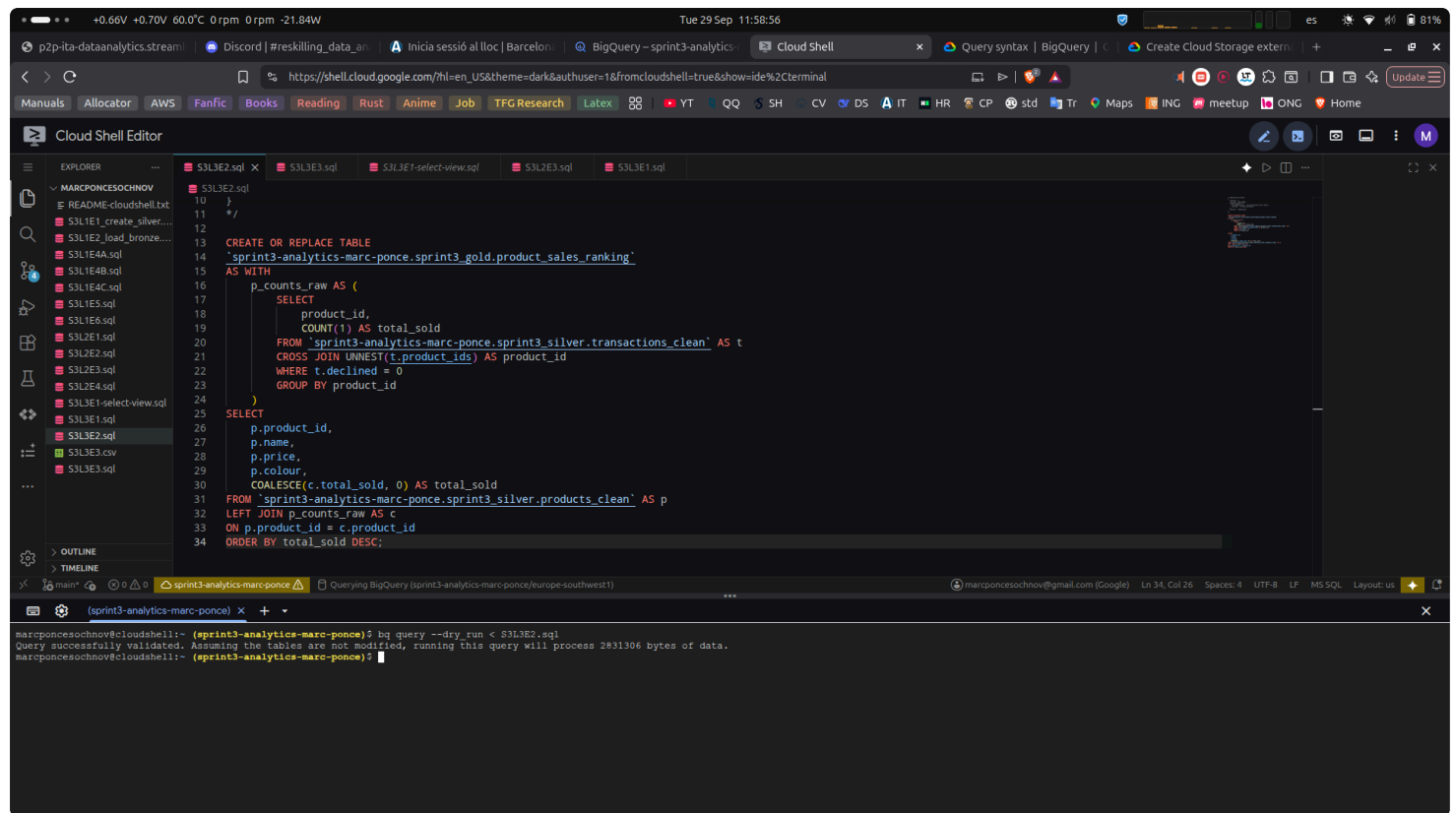
Pista Tècnica:

- A transactions_clean , els productes estan "encapsulats" dins d'un ARRAY per cada transacció.
- Primer necessitaràs "aplanar" la taula de transaccions usant UNNEST(product_ids) per poder comptar els productes individualment.
- Després, fes un encreuament (LEFT JOIN) començant per la taula de productes per no perdre aquells que no s'han venut mai.

```
CREATE OR REPLACE TABLE
`sprint3-analytics-marc-ponce.sprint3_gold.product_sales_ranking`
AS WITH
  p_counts_raw AS (
    SELECT
      product_id,
      COUNT(1) AS total_sold
    FROM `sprint3-analytics-marc-ponce.sprint3_silver.transactions_clean` AS t
    CROSS JOIN UNNEST(t.product_ids) AS product_id
    WHERE t.declined = 0
    GROUP BY product_id
  )
SELECT
  p.product_id,
  p.name,
  p.price,
  p.colour,
  COALESCE(c.total_sold, 0) AS total_sold
FROM `sprint3-analytics-marc-ponce.sprint3_silver.products_clean` AS p
LEFT JOIN p_counts_raw AS c
ON p.product_id = c.product_id
ORDER BY total_sold DESC;
```

» As it was said earlier UNNEST(...) is used to convert the ARRAY into a virtual table, which we can then JOIN LATERAL with its origin table transactions_clean . BigQuery has LATERAL reserved for

something else, so the syntax is `CROSS JOIN`. `COALESCE(val0, val1, ...)` yields the first `NOT NULL` value. In this case it is used so the `NULL` values left by the `LEFT JOIN` fall back to `0`.

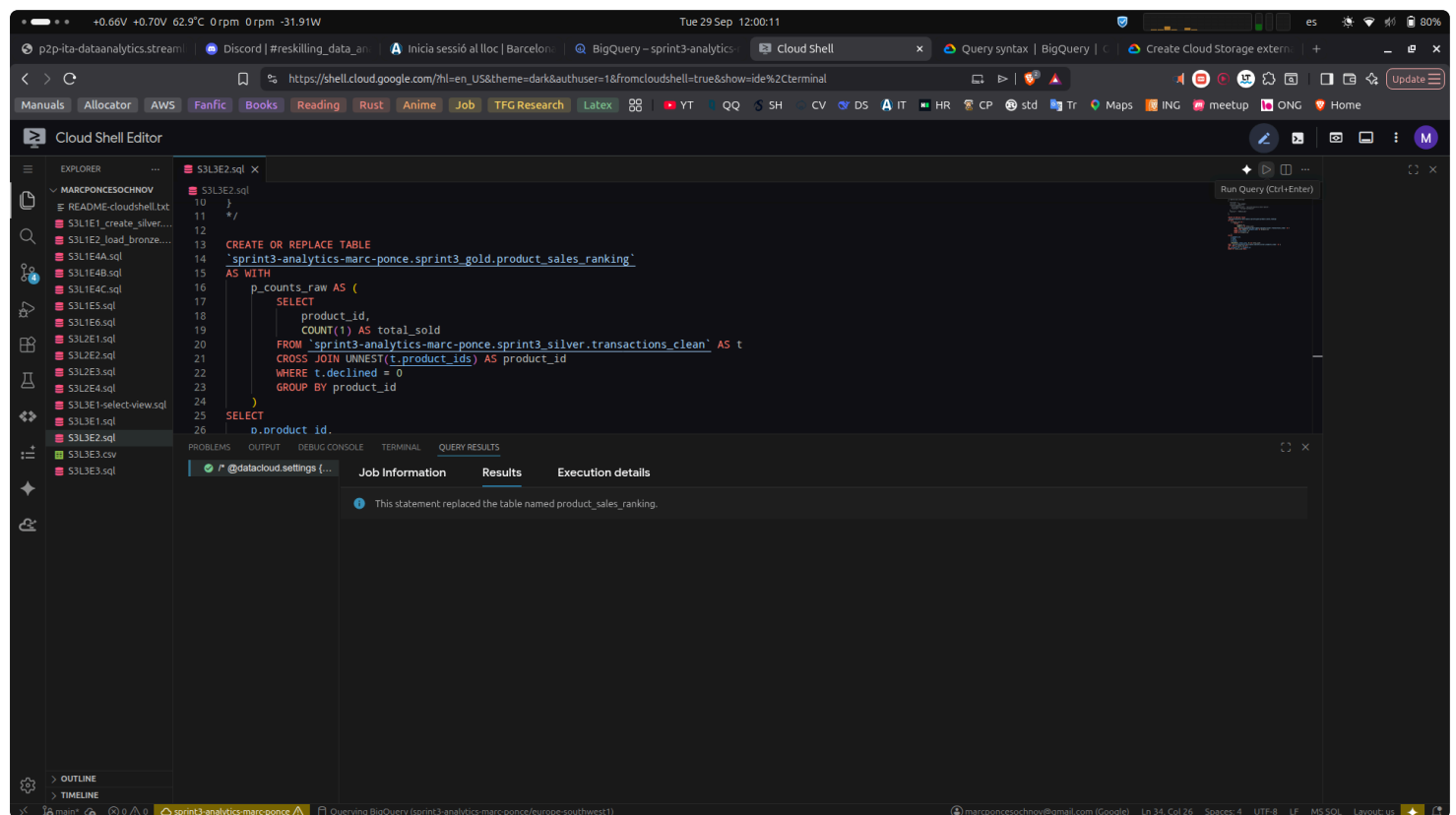


The screenshot shows the Google Cloud Shell Editor interface. The Explorer pane on the left lists files like `README-cloudshell.txt`, `S3L1E1_create_silver...`, `S3L1E2_load_bronze...`, and various `S3L1E4` through `S3L3E3` files. The main editor displays a SQL query in `S3L3E2.sql`:

```
10 }
11 */
12
13 CREATE OR REPLACE TABLE
14 `sprint3-analytics-marc-ponce.sprint3_gold.product_sales_ranking`
15 AS WITH
16   p_counts_raw AS (
17     SELECT
18       product_id,
19       COUNT(1) AS total_sold
20     FROM `sprint3-analytics-marc-ponce.sprint3_silver.transactions_clean` AS t
21     CROSS JOIN UNNEST(t.product_ids) AS product_id
22     WHERE t.declined = 0
23     GROUP BY product_id
24   )
25   SELECT
26     p.product_id,
27     p.name,
28     p.price,
29     p.colour,
30     COALESCE(c.total_sold, 0) AS total_sold
31   FROM `sprint3-analytics-marc-ponce.sprint3_silver.products_clean` AS p
32   LEFT JOIN p_counts_raw AS c
33   ON p.product_id = c.product_id
34   ORDER BY total_sold DESC;
```

The terminal at the bottom shows the execution of the query:

```
marcponcesochnov@cloudshell:~$ bq query --dry_run < S3L3E2.sql
Query successfully validated. Assuming the tables are not modified, running this query will process 283106 bytes of data.
marcponcesochnov@cloudshell:~$ (sprint3-analytics-marc-ponce)
(sprint3-analytics-marc-ponce) >
```



This screenshot shows the same Google Cloud Shell Editor interface, but with the query results displayed. The SQL query is identical to the one in the previous screenshot. The terminal shows the command `bq query` being executed. Below the terminal, the 'Query Results' pane is open, showing a message: 'This statement replaced the table named product_sales_ranking.'

The interface also shows a 'Run Query (Ctrl+Enter)' button in the top right corner of the editor area. The bottom status bar indicates the query is running on 'Europe-southwest1'.

Cloud Shell Editor interface showing a SQL query in the editor and its execution details in the results pane.

```
CREATE OR REPLACE TABLE
`sprint3-analytics-marc-ponce.sprint3_gold.product_sales_ranking`
AS WITH
  p_counts_raw AS (
    SELECT
      product_id,
      COUNT(1) AS total_sold
    FROM `sprint3-analytics-marc-ponce.sprint3_silver.transactions_clean` AS t
    CROSS JOIN UNNEST(t.product_ids) AS product_id
    WHERE t.declined = 0
    GROUP BY product_id
  )
SELECT
  p.product_id,
```

Job ID	sprint3-analytics-marc-ponce:europa-southwest1 Job_Qe0Lomf0qjEV8zKxkZDsk22TtQ
User	MarcPonceSochnov@gmail.com
Location	europa-southwest1
Creation time	Sep 29, 2026, 12:00:07 PM
Start time	Sep 29, 2026, 12:00:08 PM
End time	Sep 29, 2026, 12:00:08 PM
Duration	0.773 sec
Bytes processed	2831306
Bytes billed	20971520
Slot milliseconds	1849
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce:sprint3_gold.product_sales_ranking
Query	/* @datacloud.settings { "version": 1, "service": "BIG ...
Labels	

Exercici 3: Exportació de Resultats

El teu màner no té accés a BigQuery i vol el llistat de "Top Productes" en un Excel per a una reunió.

Tasca:

Exporta les dades de la taula `product_sales_ranking` a **Google Sheets** o baixa l'arxiu **CSV** localment.

Entrega:

Una captura de pantalla on es vegi l'arxiu obert (Excel/Sheets) amb les dades correctament formatades.

```
SELECT *
FROM `sprint3-analytics-marc-ponce.sprint3_gold.product_sales_ranking`;
```

Mon 28 Sep 13:54:50

Organización Data Analy | Discord | #reskilling_data | nou_data26: Tasca S2.01 | BigQuery - sprint3-analy | cloud shell

https://shell.cloud.google.com/console#_=_US&theme=dark&authuser=1&fromcloudshell=true&show=ide%2Cterminal

Manuales | Allocator | AWS | Fanfic | Books | Reading | Rust | Anime | Job | TFG Research | Latex | YT | QQ | SH | CV | DS | IT | HR | CP | std | Tr | Maps | ING | meetup | ONG | Home

Cloud Shell Editor

EXPLORER

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 - S3L1E2_load_bronze...
 - S3L1E4A.sql
 - S3L1E4B.sql
 - S3L1E4C.sql
 - S3L1E5.sql
 - S3L1E6.sql
 - S3L2E1.sql
 - S3L2E2.sql
 - S3L2E3.sql
 - S3L2E4.sql
 - S3L3E1-select-view.sql
 - S3L3E1.sql
 - S3L3E2.sql
 - S3L3E3.sql

S3L3E3.sql

```
2 {
3   "version": 1,
4   "service": "BIG_QUERY",
5   "connectionInfo": {
6     "billingProjectId": "sprint3-analytics-marc-ponce",
7     "location": "europe-southwest1"
8   },
9   "dialect": "GOOGLE_SQL"
10 }
11 */
12
13 SELECT *
14 FROM `sprint3-analytics-marc-ponce.sprint3_gold.product_sales_ranking`;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL QUERY RESULTS

Job Information Results Visualization Execution details

Save results

Row	product_id	name	price	colour	total_sold
1	52	riverlands the duel	31.52	#727272	2654
2	29	Tully maester Tarly	167.2	#111111	2635
3	21	duel Direwolf	96.9	#e2e2e2	2609
4	16	the duel warden	180.91	#666666	2608
5	66	mustafar jinn	2.12	#545454	2601
6	87	sith Jade	96.26	#e2e2e2	2598
7	48	rock Renly in	21.53	#9e9e9e	2597
8	99	the duel warden	127.09	#101010	2597

Items per page: 50 1 - 50 of 100

main* sprint3-analytics-marc-ponce Querying BigQuery (sprint3-analytics-marc-ponce/europe-southwest1) marcpocesochnov@gmail.com (Google) Ln 14, Col 72 Spaces: 4 UTF-8 LF MS SQL Layout: es

```
bq query \  
--format=csv \  
--max_rows=1000000 \  
< S3L3E3.sql \  
> S3L3E3.csv
```

Tue 29 Sep 10:28:27

share.streamlit.io | Discord | #reskilling_data | Inicia sessió al llor | Barce | BigQuery - sprint3-analy | cloud shell

https://shell.cloud.google.com/console#_=_US&theme=dark&authuser=1&fromcloudshell=true&show=ide%2Cterminal

Manuales | Allocator | AWS | Fanfic | Books | Reading | Rust | Anime | Job | TFG Research | Latex | YT | QQ | SH | CV | DS | IT | HR | CP | std | Tr | Maps | ING | meetup | ONG | Home

Cloud Shell Editor

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 - S3L1E2_load_bronze...
 - S3L1E4A.sql
 - S3L1E4B.sql
 - S3L1E4C.sql
 - S3L1E5.sql
 - S3L1E6.sql
 - S3L2E1.sql
 - S3L2E2.sql
 - S3L2E3.sql
 - S3L2E4.sql
 - S3L3E1-select-view.sql
 - S3L3E1.sql
 - S3L3E2.sql
 - S3L3E3.csv
 - S3L3E3.sql

S3L3E3.csv

```
1 product_id,name,price,colour,total_sold
2 52,riverlands the duel,31.52,#727272,2654
3 29,Tully maester Tarly,167.2,#111111,2635
4 21,duel Direwolf,96.9,#e2e2e2,2609
5 16,the duel warden,180.91,#666666,2608
6 66,mustafar jinn,2.12,#545454,2601
7 87,sith Jade,96.26,#e2e2e2,2598
8 48,rock Renly in,21.53,#9e9e9e,2597
9 33,duel warden,127.09,#191919,2597
10 23,riverlands north,169.96,#545454,2593
11 68,Stark Karstark,167.37,#333333,2589
12 88,Stannis warden south,167.15,#515151,2587
13 28,cheWBacca mustafar,127.44,#efefef,2584
14 4,warden south duel,171.89,#111111,2584
15 73,Dorne bastard,114.09,#848484,2584
16 31,Lannister,85.02,#f3f3f3,2580
17 34,of north,120.48,#919191,2580
18 38,ewok sith,55.43,#666666,2578
19 36,Lannister,96.22,#a5a5a5,2575
20 61,Winterfell Lannister,28.01,#848484,2575
21 54,dooku solo,13.14,#e5e5e5,2575
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL QUERY RESULTS

main* sprint3-analytics-marc-ponce

```
marcpocesochnov@cloudshell:~$ (sprint3-analytics-marc-ponce)$ bq query \  
--format=csv \  
--max_rows=1000000 \  
< S3L3E3.sql \  
> S3L3E3.csv  
marcpocesochnov@cloudshell:~$ (sprint3-analytics-marc-ponce)$
```

